Block practice · Food molecules, enzymes and moving substances

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Define diffusion, osmosis and active transport

Recall the elements and building blocks of food molecules

Describe how to carry out a food test, including DCPIP for vitamin C

Name the food-test reagent or read food-test results

Calculate area, volume and surface area to volume ratio

Explain diffusion in a context and factors affecting its rate

Explain how inhibitors or a mutation change enzyme activity

Explain how temperature or pH changes enzyme activity

Read an enzyme optimum from data and judge how reliable it is

Explain water movement by osmosis in body contexts

OLevels.com SCHOOL

Practice workbook · Define diffusion, osmosis and active transport

Definitions of the three transport processes, as written answers, matching lines, cloze passages or describing active transport.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-MEMBRANE-TRANSPORT

1. Question 2 · part b · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part b only.

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2 Molecules move in and out of cells by three methods.

These three methods are:

- diffusion
- active transport
- osmosis

(a) Give a definition of diffusion.

(2)

(b) Give a definition of active transport.

(3)

(c) Give a definition of osmosis.

(2)

(Total for Question 2 = 7 marks)

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1. Question 2 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 5 · part a · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

5 Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

(2)

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

(5)

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OLevels.com SCHOOL

2. Question 5 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part a.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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3. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 2 · part a · 2 marks

November 2021 Paper 1

Further practice · Attempt part a only.

2 Molecules move in and out of cells by three methods.

These three methods are:

- diffusion
- active transport
- osmosis

(a) Give a definition of diffusion.

(2)

(b) Give a definition of active transport.

(3)

(c) Give a definition of osmosis.

(2)

(Total for Question 2 = 7 marks)

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4. Question 2 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 2 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

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2 Molecules move in and out of cells by three methods.

These three methods are:

- diffusion
- active transport
- osmosis

(a) Give a definition of diffusion.

(2)

(b) Give a definition of active transport.

(3)

(c) Give a definition of osmosis.

(2)

(Total for Question 2 = 7 marks)

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5. Question 2 · part c · 2 marks · Working and answer

Working and answer

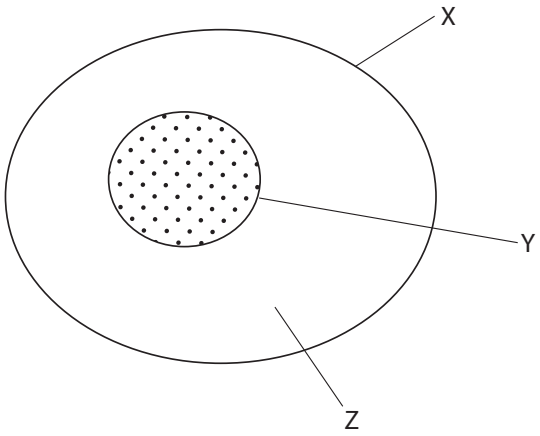
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6. Question 1 · part b · 3 marks

June 2022 Paper 1R

Further practice · Attempt part b only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

.....

(iii) Explain how body cells are organised to form organs.

(2)

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6. Question 1 · part b · 3 marks

June 2022 Paper 1R

Further practice · Attempt part b only.

(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

(3)

Process	Definition
	requires energy to move molecules against a concentration gradient
osmosis	
	water moves from a high water potential to a low water potential
diffusion	
	water moves from a low water potential to a high water potential
active transport	
	molecules move down a concentration gradient

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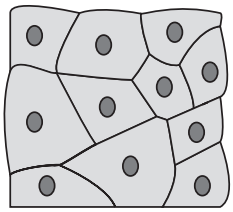
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6. Question 1 · part b · 3 marks

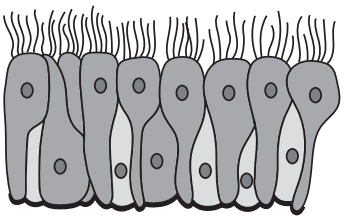
June 2022 Paper 1R

Further practice · Attempt part b only.

(c) (i) The diagram shows two types of epithelial tissue, A and B.



Tissue A



Tissue B

(Source: © Timonina/Shutterstock)

Give the name of each type of epithelial tissue.

(2)

A .....

B .....

(ii) An epithelial cell has an actual width of 60 micrometres (μm).

The cell is magnified 40 times.

Calculate the image width of this cell in mm.

[1 mm = 1000 μm]

(3)

image width = ... mm

(Total for Question 1 = 14 marks)

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6. Question 1 · part b · 3 marks · Working and answer

Working and answer

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7. Question 3 · part a.iii · 1 mark

January 2023 Paper 2

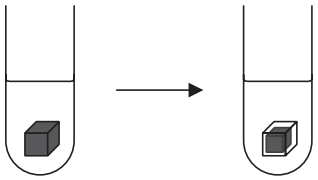
Further practice · Attempt part a.iii only.

3 A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

(3)

surface area to volume ratio = ...

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

(1)

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

(1)

.....

.....

.....

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7. Question 3 · part a.iii · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.iii only.

(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

(2)

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

(i) Describe a method for this investigation using the apparatus shown in the diagram.

(2)

(ii) Give two variables the student would need to control in their investigation.

(2)

1 .....

2 .....

(Total for Question 3 = 11 marks)

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7. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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8. Question 1 · part whole · 6 marks

June 2023 Paper 1

Further practice · Attempt part whole only.

1 Use information from the box to complete the sentences about the movement of molecules.

(6)

ADP	ATP	against	along	down	fully
gases	selectively	volume	water		

Molecules can move in and out of cells by osmosis, diffusion and active transport.

Osmosis is the movement of ..... across a ..... permeable membrane.

Diffusion allows the passage of molecules ..... a concentration gradient. The speed of movement is affected by the surface area to ..... ratio of the cell.

Active transport requires the use of energy stored in the form of ..... and allows molecules to move ..... a concentration gradient.

(Total for Question 1 = 6 marks)

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8. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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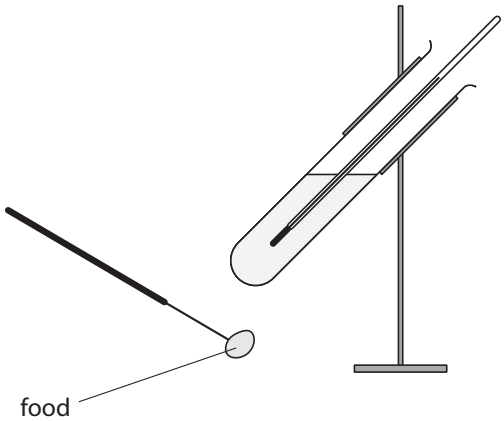
9. Question 7 · part b · 2 marks

November 2024 Paper 1

Further practice · Attempt part b only.

7 In a healthy person, the amount of energy in the daily diet should equal the amount of energy used by the body.

(a) The equipment shown can be used to measure the amount of energy in food.



Describe how this equipment could be used to compare the energy content of different foods.

(6)

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9. Question 7 · part b · 2 marks

November 2024 Paper 1

Further practice · Attempt part b only.

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(b) Describe a named method of transport that uses energy to move molecules across a cell membrane.

(2)

.....

.....

.....

.....

(Total for Question 7 = 8 marks)

Working and answer

OLevels.COM SCHOOL

Practice workbook · Recall the elements and building blocks of food molecules

Chemical make-up of carbohydrates, proteins and lipids: their elements and the small units that join to build them.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-NUTRIENT-COMPOSITION-RECALL

1. Question 3 · part a.i · 1 mark

June 2021 Paper 1

Worked-example retry · Attempt part a.i only.

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3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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1. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part a · 3 marks

June 2019 Paper 2

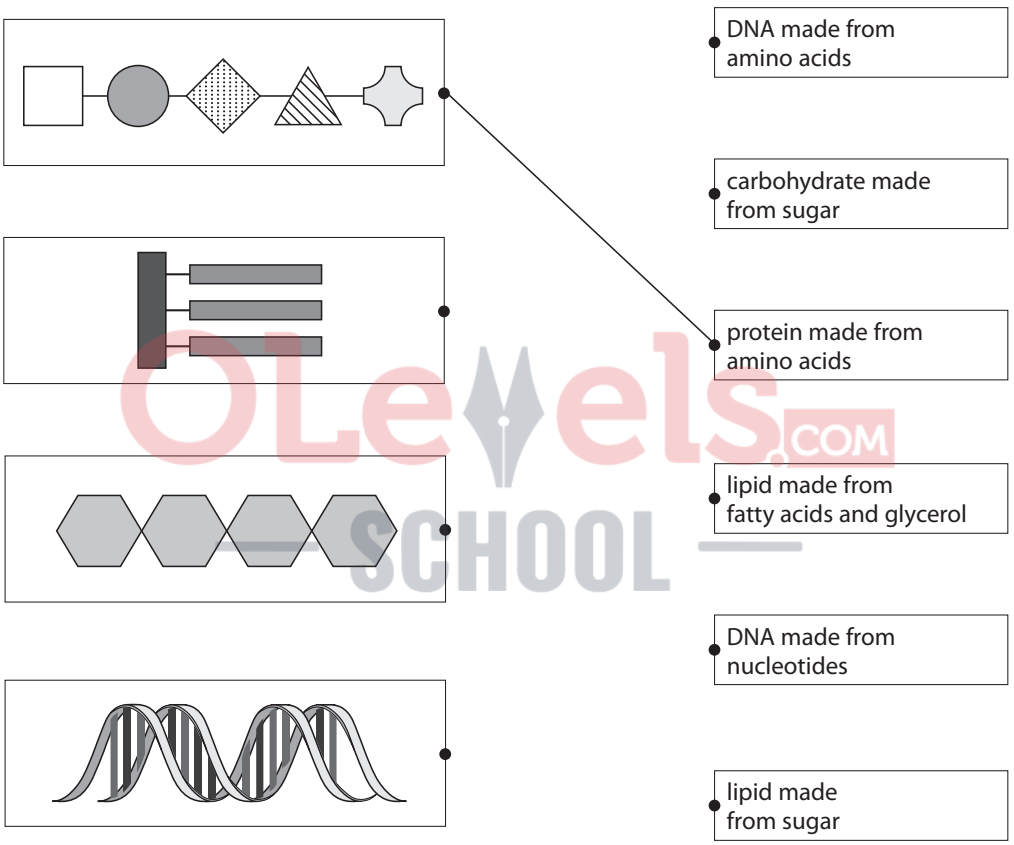
Further practice · Attempt part a only.

3 (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

(3)



(b) Which body organ produces enzymes that break down protein?

(1)

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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P 5 8 5 6 6 A 0 6 2 8

2. Question 3 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 2 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

2 A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

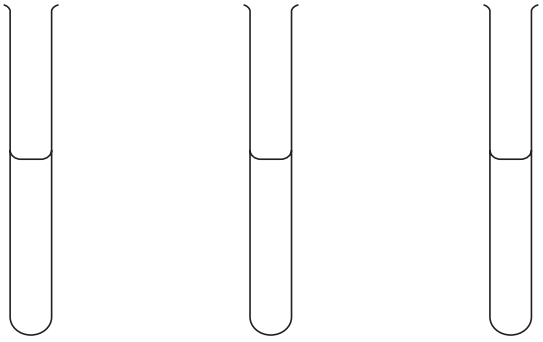


Table 1 lists the contents of each test tube.

Test tube A	Test tube B	Test tube C
5 cm^3 of milk	5 cm^3 of milk	5 cm^3 of milk
2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate
6 drops of phenolphthalein	6 drops of phenolphthalein	6 drops of phenolphthalein
1 cm^3 enzyme	1 cm^3 enzyme	1 cm^3 boiled enzyme
distilled water	bile salts	bile salts

Table 1

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3. Question 2 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

	Colour of contents		
	Tube A	**Tube B**	**Tube C**
at start	pink	pink	pink
after 5 minutes	pink	colourless	pink
after 10 minutes	pink	colourless	pink
after 15 minutes	colourless	colourless	pink

Table 2

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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3. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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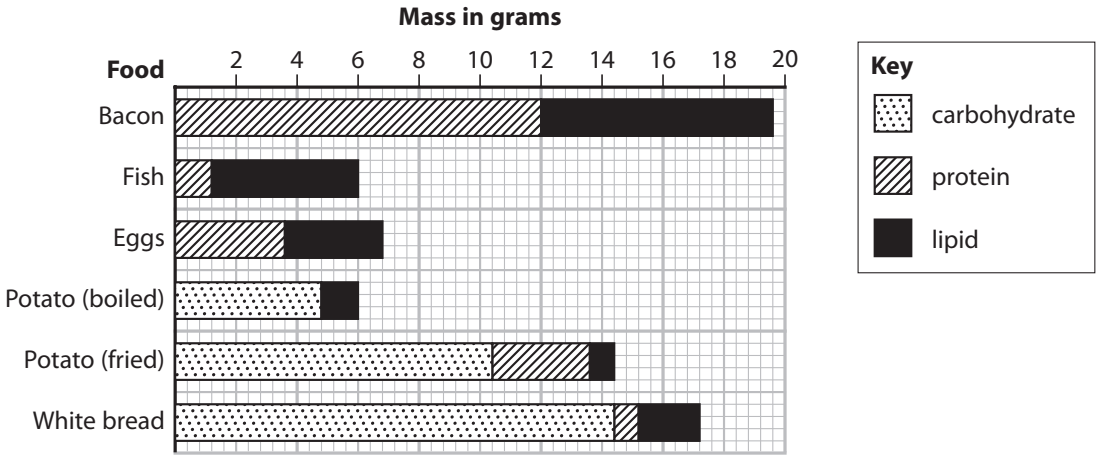
4. Question 7 · part a.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part a.i only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

.....

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

.....

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

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4. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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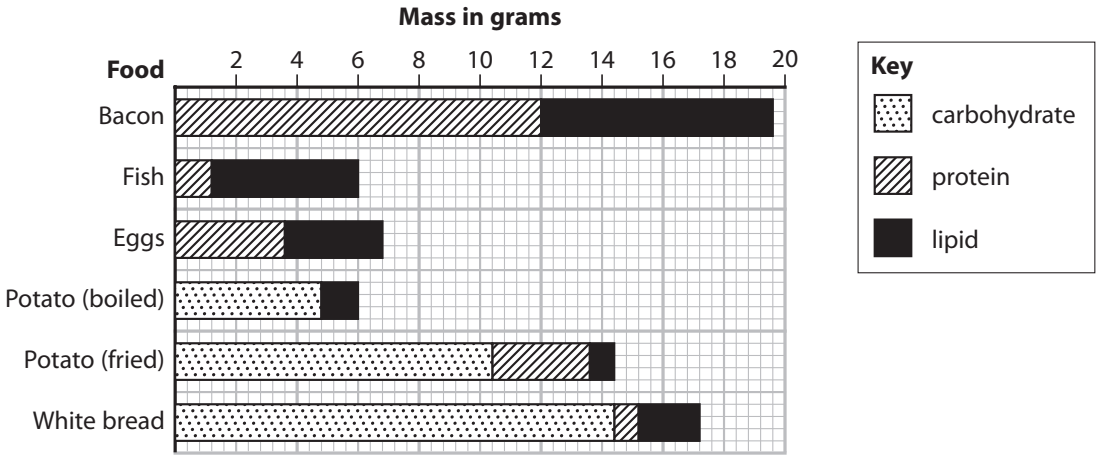
5. Question 7 · part a.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part a.ii only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

.....

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

.....

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

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P 6 7 0 5 9 A 0 2 0 2 4

5. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part a · 2 marks

June 2022 Paper 2R

Further practice · Attempt part a only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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6. Question 5 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part a.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

.....

.....

.....

.....

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

.....

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

.....

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25 to 29.9	overweight
30 to 34.9	obese
35 to 39.9	severely obese
more than 39.9	morbidly obese

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7. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Describe how to carry out a food test, including DCPIP for vitamin C

Writing the method for Benedict's, iodine, biuret, emulsion or DCPIP tests on foods, urine, plasma or tubing contents, including safe conduct and a DCPIP comparison of two juices.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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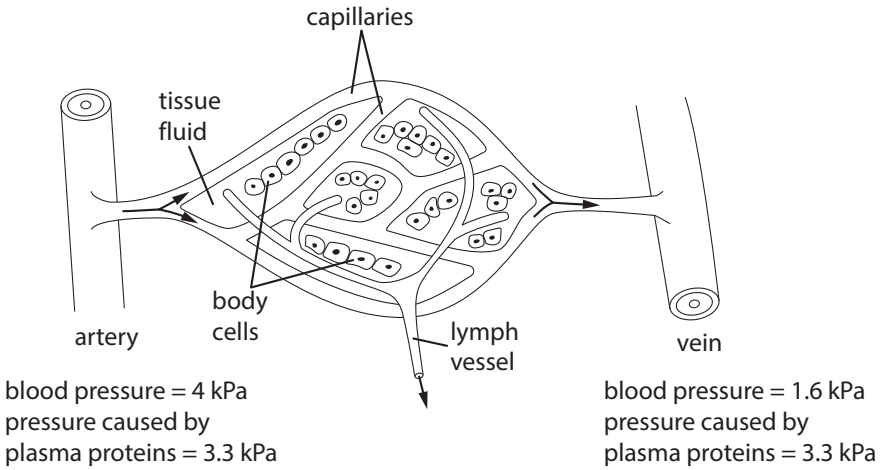
Type ID: HBI-FOOD-TEST-PROCEDURE

1. Question 7 · part b · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

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1. Question 7 · part b · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

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(b) Describe how a sample of plasma could be tested for the presence of proteins.

(4)

(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

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1. Question 7 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part c · 3 marks

January 2020 Paper 1

Further practice · Attempt part c only.

1 (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.



(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

(1)

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

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2. Question 1 · part c · 3 marks

January 2020 Paper 1

Further practice · Attempt part c only.

(b) The diagram shows the nutritional label on a package containing yogurt.

Nutrition Facts

Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

Amount/Serving	% Daily Value*	Amount/Serving	% Daily Value*
Total Fat	a%	Sodium	2%
Saturated Fat	0%	Total Carb.	6%
Trans Fat		Sugars	
Cholesterol	1%	Protein	22%

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

(1)

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

(1)

- ☒ **A** it contains vitamins C and D
- ☒ **B** it contains less than 5 mg of cholesterol
- ☒ **C** it does not contain energy from carbohydrates
- ☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

(3)

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(Total for Question 1 = 6 marks)

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OLevels.com SCHOOL

2. Question 1 · part c · 3 marks · Working and answer

Working and answer

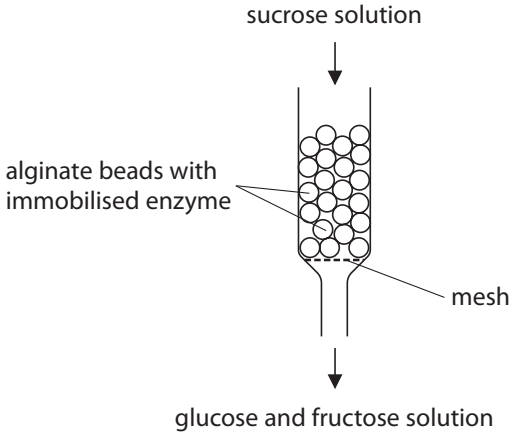
OLevels.com SCHOOL

3. Question 6 · part a.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

1 .....

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2 .....

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3. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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4. Question 7 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

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4. Question 7 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

Number of bubbles of CO_2	Time taken for chymosin to turn milk protein solid, in seconds				
	Test 1	Test 2	Test 3	Test 4	Mean
100	253	257	250	260	255
150	238	232	241	229	
200	216	214	219	211	215
250	208	202	212	198	205
300	210	200	199	311	203

(i) State three factors that the student should control.

(3)

1 .....

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2 .....

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3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

(2)

mean time taken = ...

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P 6 2 0 5 4 A 0 2 0 2 4

4. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

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5. Question 3 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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5. Question 3 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

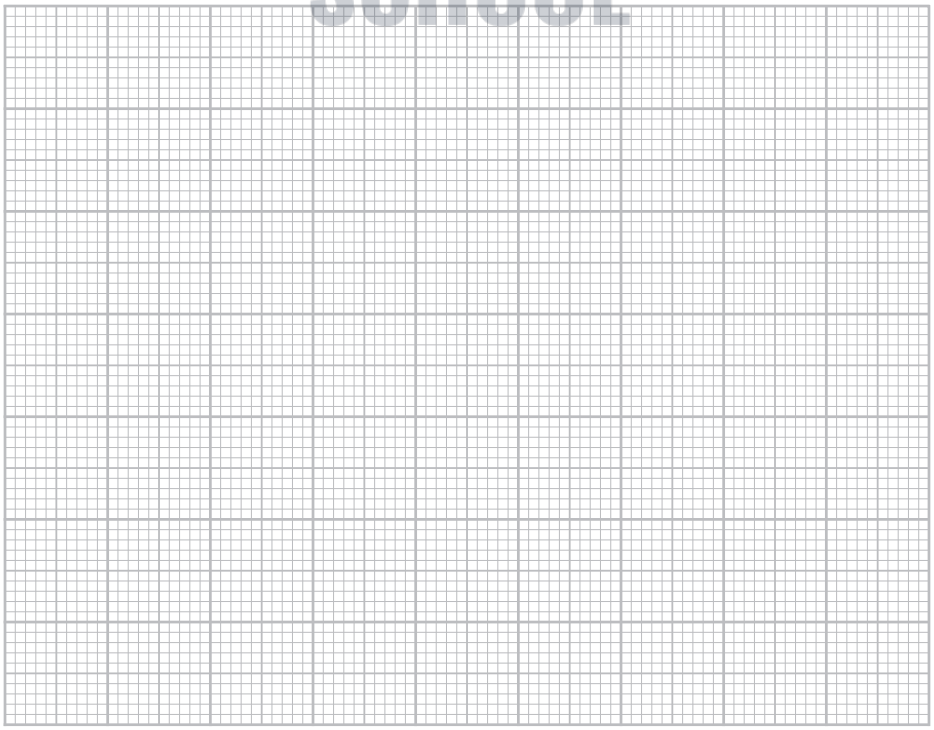
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



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P 6 6 4 3 2 A 0 8 2 4

5. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 2 · part b · 3 marks

June 2023 Paper 2

Further practice · Attempt part b only.

2 The table gives the nutritional information for a pot of yoghurt.

	Mass per 100 g of yoghurt	Mass per 85 g pot
fat	2.7 g	2.3 g
carbohydrate	12.6 g	10.7 g
protein	6.1 g	
calcium	149 mg	127 mg
vitamin D	0.9 µg	0.8 µg

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

(3)

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mass of protein = ... g

(b) Describe how the yoghurt could be tested to show that protein is present.

(3)

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6. Question 2 · part b · 3 marks

June 2023 Paper 2

Further practice · Attempt part b only.

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(c) A pot of yoghurt provides 16% of a person's daily requirement of calcium.

Calculate how many pots of yoghurt a person would need to eat to obtain their daily requirement of calcium.

(2)

number of pots = ...

(d) Explain why yoghurt is good for bone development.

(5)

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(Total for Question 2 = 13 marks)

6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

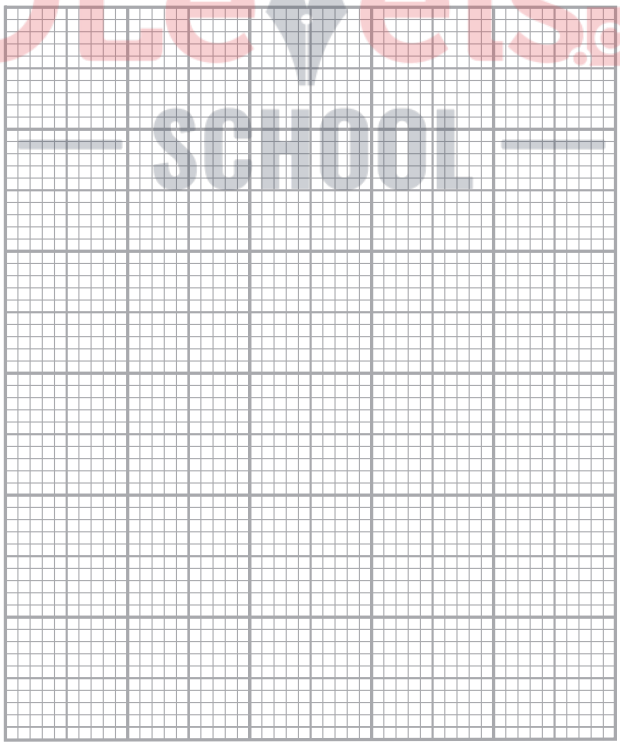
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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7. Question 4 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

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(vi) Name a part of the body where starch digestion takes place.

(1)

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(b) Describe a test for starch in a sample of food.

(2)

(Total for Question 4 = 13 marks)

7. Question 4 · part b · 2 marks · Working and answer

Working and answer

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8. Question 2 · part b.iv · 5 marks

November 2024 Paper 2

Further practice · Attempt part b.iv only.

2 (a) Describe what is meant by a balanced diet.

(2)

.....

.....

.....

.....

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

.....

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8. Question 2 · part b.iv · 5 marks

November 2024 Paper 2

Further practice · Attempt part b.iv only.

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(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

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number of slices = ...

P 7 5 9 5 9 A 0 5 2 4

8. Question 2 · part b.iv · 5 marks

November 2024 Paper 2

Further practice · Attempt part b.iv only.

(iv) When meat is rubbed on a piece of paper it leaves a greasy mark due to the presence of fat in the meat.

Describe another test to show the presence of fat in the meat.

(5)

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(Total for Question 2 = 13 marks)

8. Question 2 · part b.iv · 5 marks · Working and answer

Working and answer

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9. Question 2 · part c.i · 5 marks

June 2019 Paper 1R

Further practice · Attempt part c.i only.

2 Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

(2)

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

(2)

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9. Question 2 · part c.i · 5 marks

June 2019 Paper 1R

Further practice · Attempt part c.i only.

(c) The vitamin C content of fruit juices varies.

(i) Design an investigation to compare the vitamin C content of fresh fruit juice and long-life fruit juice.

(5)

(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(Total for Question 2 = 11 marks)

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9. Question 2 · part c.i · 5 marks · Working and answer

Working and answer

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10. Question 7 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

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10. Question 7 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

(iii) Calculate the mass, in grams, of boiled broccoli that should be eaten to provide the recommended daily intake of vitamin C. (lines 12 to 14)

(3)

mass = ... g

(b) DCPIP is a chemical used to determine the amount of vitamin C in food.

Describe a method, using DCPIP, to compare the amount of vitamin C in fresh orange juice with that in boiled orange juice.

(5)

(Total for Question 7 = 12 marks)

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10. Question 7 · part b · 5 marks · Working and answer

Working and answer

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11. Question 2 · part a · 4 marks

November 2021 Paper 2

Further practice · Attempt part a only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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11. Question 2 · part a · 4 marks

November 2021 Paper 2

Further practice · Attempt part a only.

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(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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11. Question 2 · part a · 4 marks · Working and answer

Working and answer

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12. Question 2 · part a.i · 4 marks

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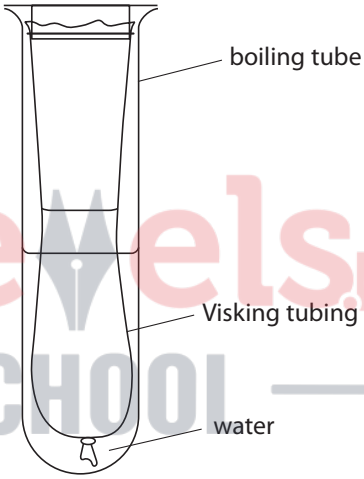
Further practice · Attempt part a.i only.

2 A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 cm^3 of starch suspension and 5 cm^3 of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.



After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

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12. Question 2 · part a.i · 4 marks

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Further practice · Attempt part a.i only.

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(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

(4)

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

(3)

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

(3)

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12. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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13. Question 2 · part a.ii · 3 marks

January 2022 Paper 1

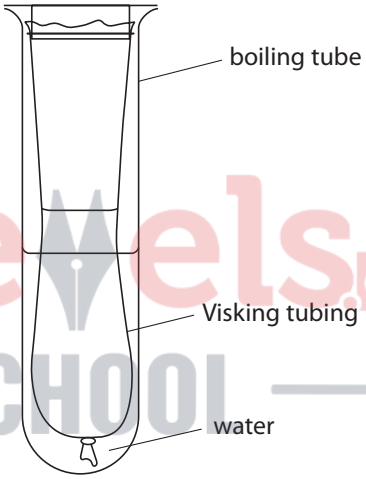
Further practice · Attempt part a.ii only.

2 A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 cm^3 of starch suspension and 5 cm^3 of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.



After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

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13. Question 2 · part a.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.ii only.

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(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

(4)

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

(3)

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

(3)

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13. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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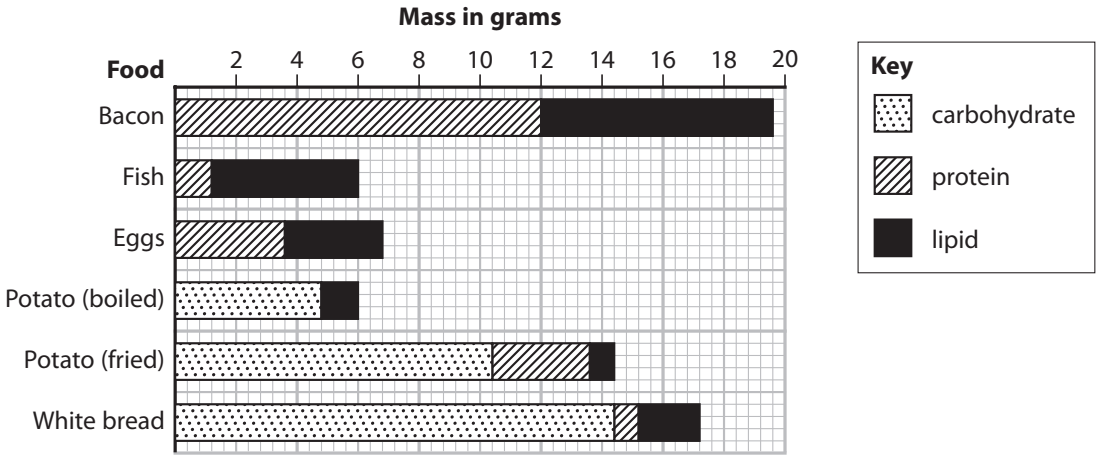
14. Question 7 · part b · 3 marks

January 2022 Paper 1

Further practice · Attempt part b only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

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P 6 7 0 5 9 A 0 2 0 2 4

14. Question 7 · part b · 3 marks

January 2022 Paper 1

Further practice · Attempt part b only.

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(iv) The energy in one gram of protein is 17 kJ and the energy in one gram of lipid is 37 kJ.

Calculate the total energy in 26 g of fish.

(3)

total energy = ... kJ

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(b) Describe how to test for the presence of lipid in the fish.

(3)

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(Total for Question 7 = 10 marks)

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14. Question 7 · part b · 3 marks · Working and answer

Working and answer

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15. Question 4 · part a.i · 3 marks

June 2022 Paper 1R

Further practice · Attempt part a.i only.

4 (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

(3)

(ii) Name the enzyme that digests lipids.

(1)

(iii) Name the products of lipid digestion.

(2)

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

Temperature at 37 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	8.5	8.0	7.5	7.2	7.0	6.8	6.2	5.8	5.5	5.5

Temperature at 80 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	9.2	9.0	8.8	8.8		8.6	8.6	8.4	8.4	8.4

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15. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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16. Question 4 · part c · 4 marks

June 2022 Paper 2

Further practice · Attempt part c only.

4 A student tests foods W, X and Y for the presence of different nutrients.

The student tests each of the foods using three different solutions, A, B and C.

The table shows the student's results.

Food	Solution used in test		
	A	B	C
W	no change	no change	blue-black
X	lilac	no change	no change
Y	no change	brick-red	no change

(a) Name solution C.

(1)

.....

(b) Name the nutrient present in food X.

(1)

.....

(c) Describe a safe method the student could use to test a food using solution B.

(4)

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(Total for Question 4 = 6 marks)

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16. Question 4 · part c · 4 marks · Working and answer

Working and answer

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17. Question 2 · part b · 5 marks

November 2023 Paper 1

Further practice · Attempt part b only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

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17. Question 2 · part b · 5 marks

November 2023 Paper 1

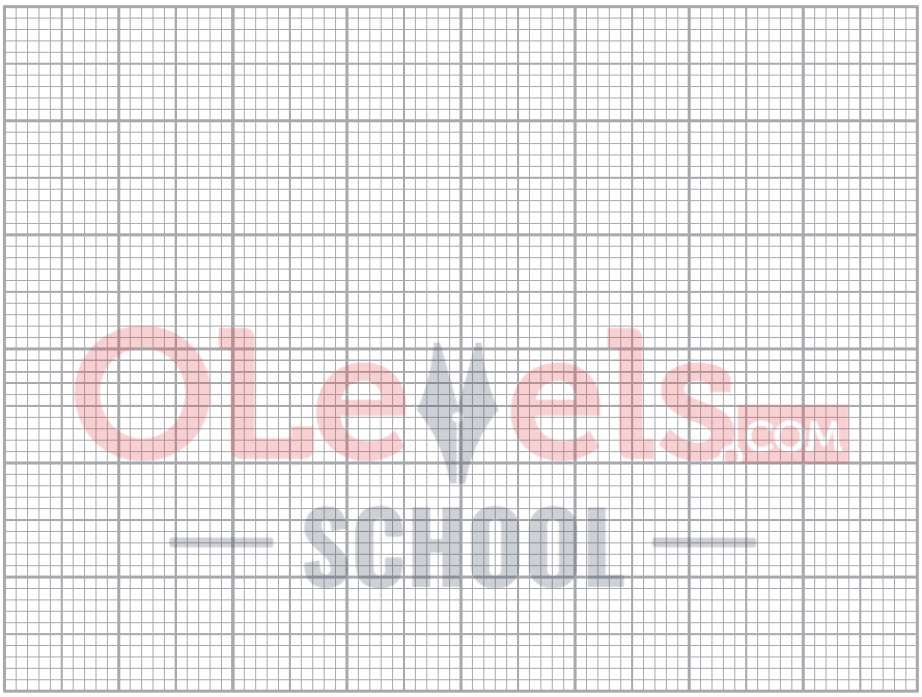
Further practice · Attempt part b only.

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(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

17. Question 2 · part b · 5 marks

November 2023 Paper 1

Further practice · Attempt part b only.

(iii) Give **three** differences between the glucose levels for the two people during the three hours.

(3)

1 .....

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2 .....

.....

3 .....

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(b) One symptom of diabetes is glucose in the urine.

Describe how a sample of urine could be tested safely for glucose.

(5)

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(Total for Question 2 = 16 marks)

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17. Question 2 · part b · 5 marks · Working and answer

Working and answer

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Practice workbook · Name the food-test reagent or read food-test results

One-step food-test questions: naming the reagent or equipment for a test, or reading colour outcomes to decide which nutrients each sample holds.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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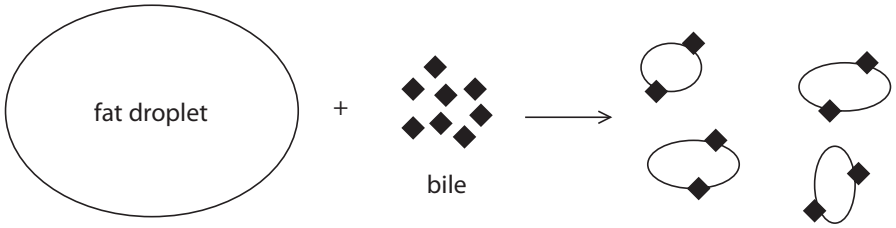
Type ID: HBI-FOOD-TEST-INTERPRET

1. Question 3 · part b.i · 3 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.i only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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1. Question 3 · part b.i · 3 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.i only.

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(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

Reagent used	Colour of the food solution after the test
iodine solution	blue/black
biuret solution	purple
Benedict's solution	blue

(i) Describe what the food tests show about the dietary components in the food.

(3)

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(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

(3)

1. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

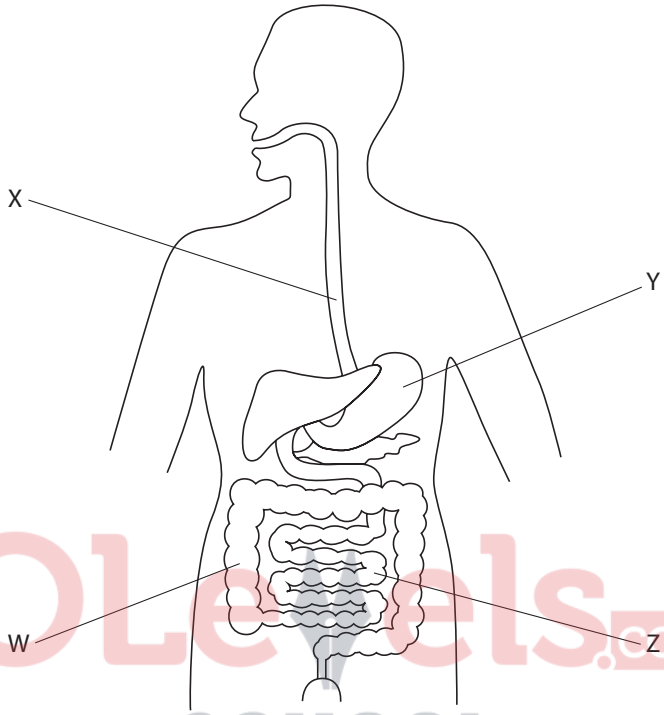
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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

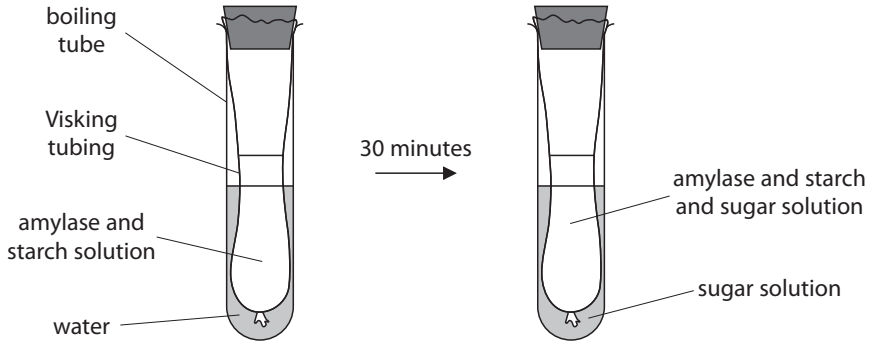
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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

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(v) After 30 minutes iodine solution is added to the contents of the Visking tubing.

Explain what the results of this test would show.

(2)

(Total for Question 2 = 15 marks)

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Working and answer

3. Question 5 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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3. Question 5 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 5 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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4. Question 5 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

(3)

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4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part c.i · 1 mark

June 2019 Paper 2

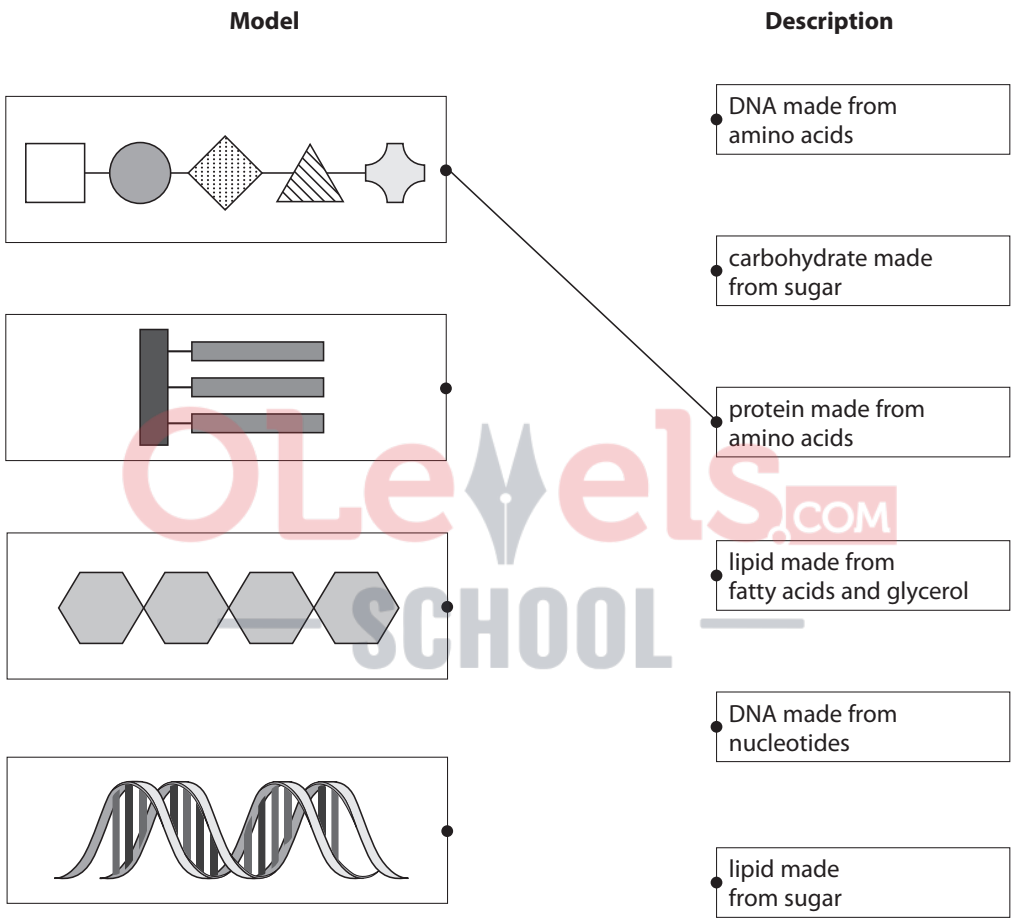
Further practice · Attempt part c.i only.

3 (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

(3)



(b) Which body organ produces enzymes that break down protein?

(1)

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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5. Question 3 · part c.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part c.i only.

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(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

Substance	Colour after Biuret test
milk	purple
pasta	blue
lemon juice	blue
cheese	purple
distilled water	purple

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(i) In the student's test, how many of the substances give a positive result for protein?

(1)

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

(1)

.....

.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

(2)

Hazard	Reducing risk
stools – trip hazard	keep stools under bench
broken glass – cuts	
Biuret reagent – irritant	

(Total for Question 3 = 8 marks)

5. Question 3 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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6. Question 2 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

.....

(ii) Identify which powders contain protein but no starch.

(2)

.....

.....

(iii) Identify which powder contains reducing sugar but no protein.

(1)

.....

(iv) Give the contents of powder W.

(1)

.....

(Total for Question 2 = 9 marks)

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Working and answer

7. Question 2 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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7. Question 2 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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Working and answer

8. Question 2 · part b.iii · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iii only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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8. Question 2 · part b.iii · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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Working and answer

9. Question 2 · part b.iv · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iv only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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9. Question 2 · part b.iv · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA

(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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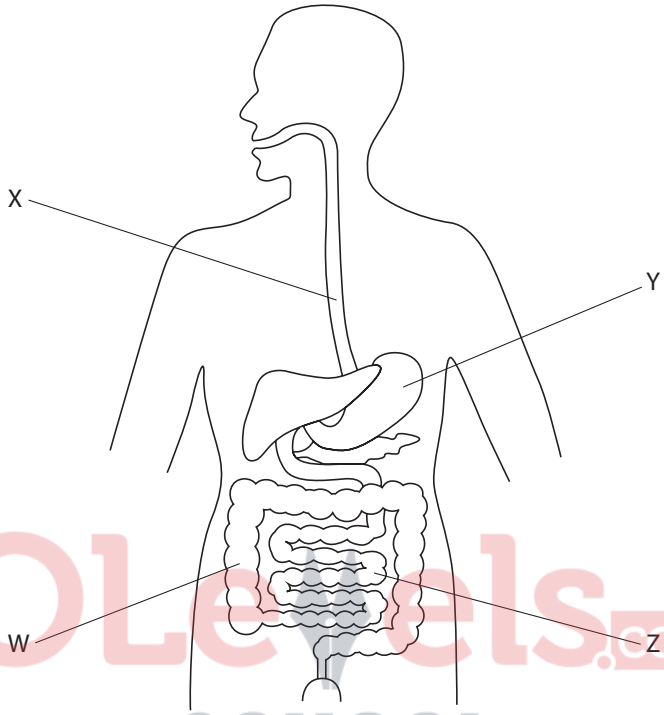
Working and answer

10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

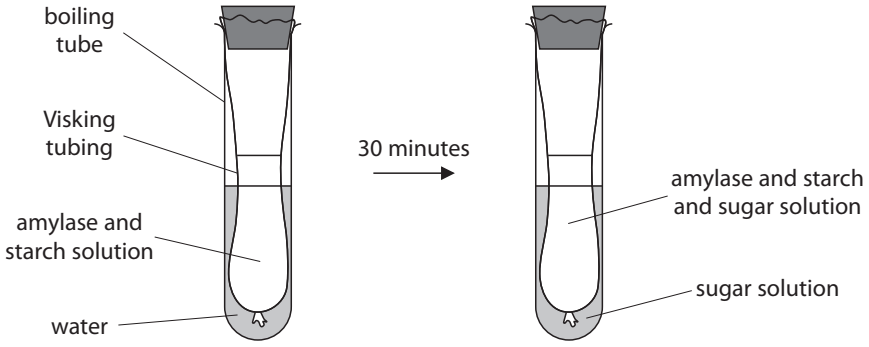
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10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

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(v) After 30 minutes iodine solution is added to the contents of the Visking tubing.

Explain what the results of this test would show.

(2)

(Total for Question 2 = 15 marks)

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Working and answer

11. Question 4 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

4 A student tests foods W, X and Y for the presence of different nutrients.

The student tests each of the foods using three different solutions, A, B and C.

The table shows the student's results.

Food	Solution used in test		
	A	B	C
W	no change	no change	blue-black
X	lilac	no change	no change
Y	no change	brick-red	no change

(a) Name solution C.

(1)

.....

(b) Name the nutrient present in food X.

(1)

.....

(c) Describe a safe method the student could use to test a food using solution B.

(4)

.....

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(Total for Question 4 = 6 marks)

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11. Question 4 · part a · 1 mark · Working and answer

Working and answer

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12. Question 4 · part b · 1 mark

June 2022 Paper 2

Further practice · Attempt part b only.

4 A student tests foods W, X and Y for the presence of different nutrients.

The student tests each of the foods using three different solutions, A, B and C.

The table shows the student's results.

Food	Solution used in test		
	A	B	C
W	no change	no change	blue-black
X	lilac	no change	no change
Y	no change	brick-red	no change

(a) Name solution C.

(1)

.....

(b) Name the nutrient present in food X.

(1)

.....

(c) Describe a safe method the student could use to test a food using solution B.

(4)

.....

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(Total for Question 4 = 6 marks)

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12. Question 4 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · Calculate area, volume and surface area to volume ratio

Cube side lengths or a measured circle are given; surface area, volume, their ratio, or a circular area is required.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-AREA-VOLUME-SA-V

1. Question 3 · part a.i · 3 marks

January 2023 Paper 2

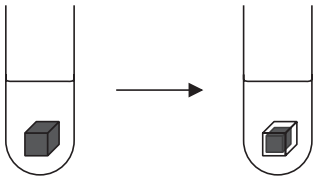
Worked-example retry · Attempt part a.i only.

3 A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

(3)

surface area to volume ratio = ...

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

(1)

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

(1)

.....

.....

.....

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1. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

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2. Question 6 · part c.i · 3 marks

June 2019 Paper 1

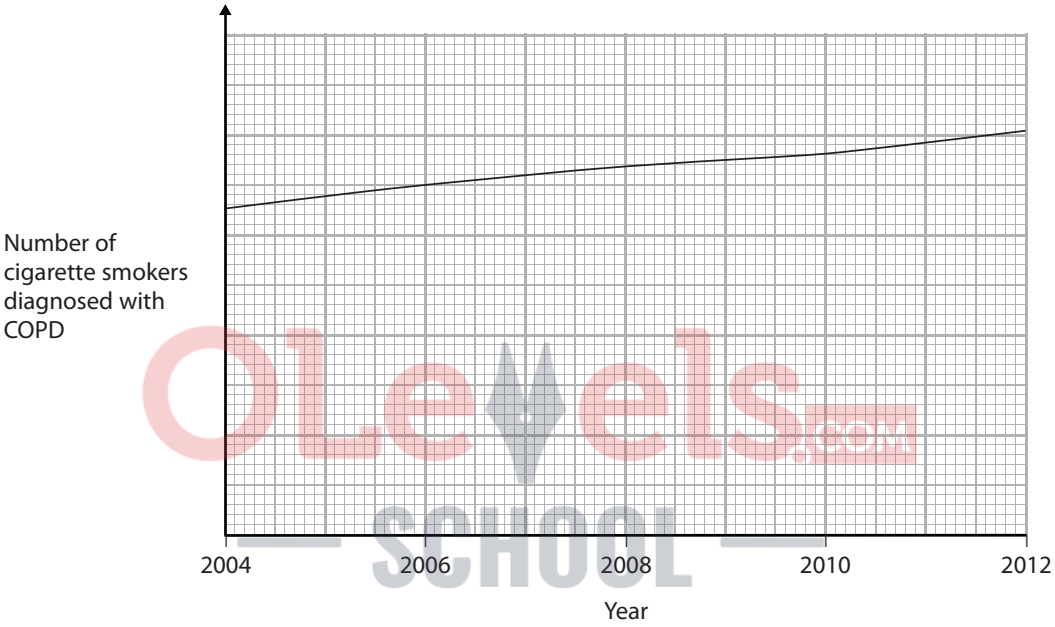
Further practice · Attempt part c.i only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

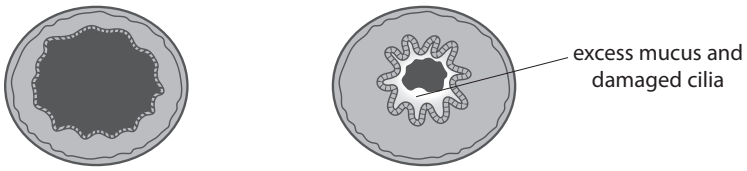
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2. Question 6 · part c.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part c.i only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.



healthy airway

chronic bronchitis

Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

(2)

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2. Question 6 · part c.i · 3 marks

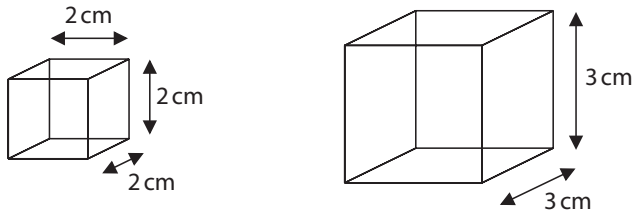
June 2019 Paper 1

Further practice · Attempt part c.i only.

(c) The diagram shows models of two alveoli.

One model represents an alveolus from a person with healthy lungs.

The other model represents an alveolus from a person with emphysema.



Healthy **Emphysema**

The table shows the surface area to volume ratio for a healthy alveolus.

	Surface area in cm^2	Volume in cm^3	Surface area to volume ratio
Healthy	24	8	3 : 1
Emphysema			

(i) Complete the table by giving the missing information.

(3)

(ii) Explain how the surface area to volume ratio of alveoli in the lungs of a person with emphysema will affect the normal function of body cells.

(3)

(Total for Question 6 = 11 marks)

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2. Question 6 · part c.i · 3 marks · Working and answer

Working and answer

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3. Question 6 · part a.i · 2 marks

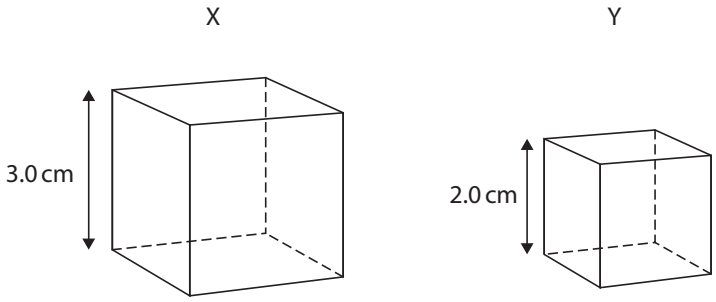
June 2024 Paper 1

Further practice · Attempt part a.i only.

6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

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3. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 6 · part a.ii · 2 marks

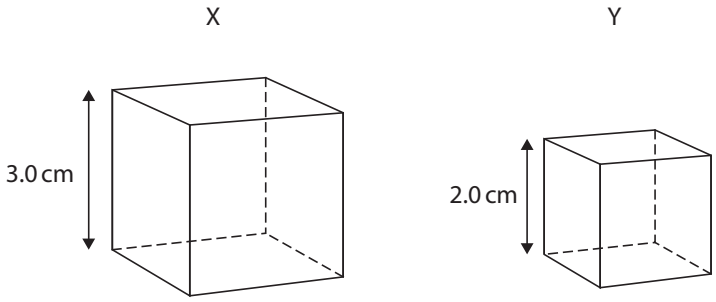
June 2024 Paper 1

Further practice · Attempt part a.ii only.

6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

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4. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 6 · part a.iii · 2 marks

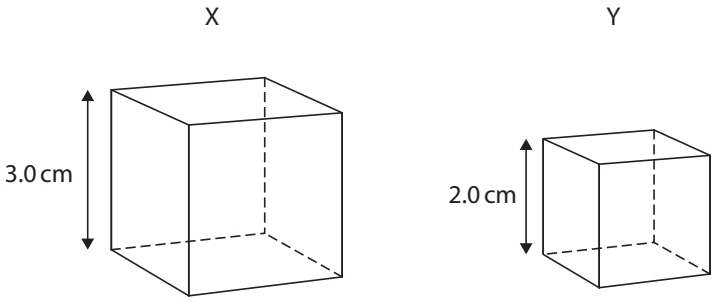
June 2024 Paper 1

Further practice · Attempt part a.iii only.

6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

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5. Question 6 · part a.iii · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.iii only.

(b) Multicellular organisms, with a small SA:V ratio, such as humans, need an efficient transport system to deliver substances to body cells.

Describe the role of the human blood transport system in delivering substances to body cells.

(5)

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5. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

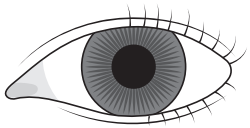
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6. Question 6 · part b · 3 marks

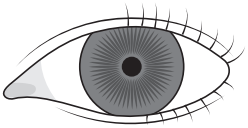
June 2024 Paper 2

Further practice · Attempt part b only.

6 The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

(3)

(b) A student calculated the area of the pupil in bright light to be 7.1 mm^2.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = πr^2 $\pi = 3.14$]

(3)

area of pupil in dim light = ... mm^2

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P 7 4 6 3 0 A 0 1 6 2 4

6. Question 6 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(c) (i) Explain how the size of the pupil is adjusted from bright light to dim light.

(3)

(ii) Explain the adjustment made to the pupil when moving into dim light.

(2)

(Total for Question 6 = 11 marks)

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6. Question 6 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain diffusion in a context and factors affecting its rate

Diffusion of gases or solutes in cells, bacteria, gut and dialysis, and how surface area to volume ratio and concentration gradient change rate.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-DIFFUSION-RATE-IN-CONTEXT

1. Question 5 · part a · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part a only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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1. Question 5 · part a · 3 marks · Working and answer

Working and answer

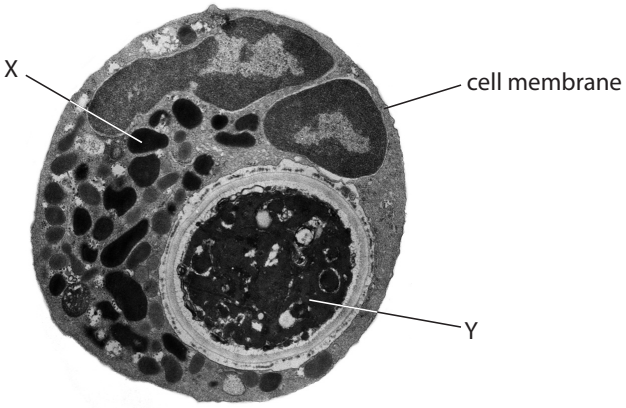
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2. Question 2 · part b · 2 marks

June 2019 Paper 2

Further practice · Attempt part b only.

2 (a) The image is an electron micrograph of a human body cell.



(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

(2)

Part	Name of part	Function
X		release energy from glucose
Y	nucleus	

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

(1)

- ☒ **A** more cell structures can be seen with greater resolution
- ☒ **B** more cell structures can be seen with less resolution
- ☒ **C** fewer cell structures can be seen with greater resolution
- ☒ **D** fewer cell structures can be seen with less resolution

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2. Question 2 · part b · 2 marks

June 2019 Paper 2

Further practice · Attempt part b only.

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(iii) The actual diameter of this human body cell before magnification is 0.05 mm.

Use information from the image to calculate the magnification of the cell.

(3)

magnification = ...

(b) Body cells produce carbon dioxide.

Describe how carbon dioxide passes from a body cell into the blood.

(2)

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(Total for Question 2 = 8 marks)

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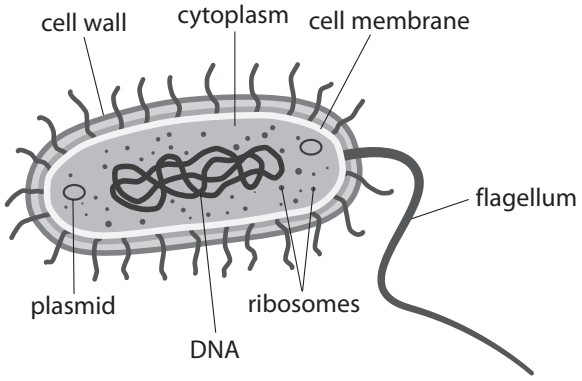
Working and answer

3. Question 3 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

3 (a) The diagram shows a type of bacterial cell.



(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

(3)

Structure	Function
cell membrane	protects against pathogens
	helps the cell to move
ribosomes	makes proteins
	holds the genetic code
flagellum	
	controls which substances enter and exit the cell

(ii) State two structural differences between the bacterial cell and a sperm cell.

(2)

1 .....

.....

2 .....

.....

8

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3. Question 3 · part b.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ... → ..... + water

(iii) Explain why respiration is important for bacteria.

(2)

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(Total for Question 3 = 10 marks)

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Working and answer

4. Question 6 · part c · 2 marks

June 2024 Paper 1

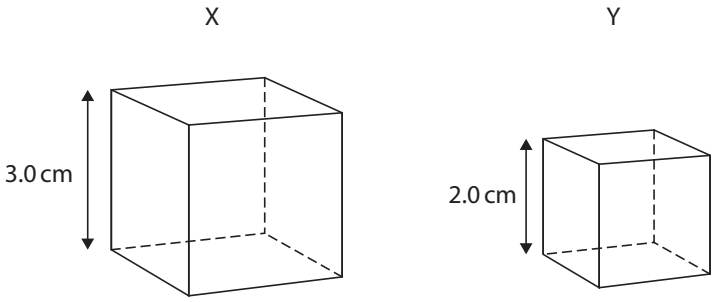
Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

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(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

4. Question 6 · part c · 2 marks

June 2024 Paper 1

Further practice · Attempt part c only.

(b) Multicellular organisms, with a small SA:V ratio, such as humans, need an efficient transport system to deliver substances to body cells.

Describe the role of the human blood transport system in delivering substances to body cells.

(5)

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4. Question 6 · part c · 2 marks

June 2024 Paper 1

Further practice · Attempt part c only.

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(c) Bacteria are able to exchange materials directly with their environment as they have a very high SA : V ratio.

They do not have an internal transport system.

Describe how bacteria exchange materials with their environment.

(2)

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(Total for Question 6 = 13 marks)

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Working and answer

5. Question 3 · part a.iv · 2 marks

January 2023 Paper 2

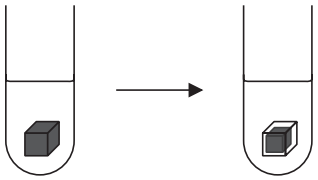
Further practice · Attempt part a.iv only.

3 A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

(3)

surface area to volume ratio = ...

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

(1)

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

(1)

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P 7 2 6 0 2 A 0 8 2 8

5. Question 3 · part a.iv · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iv only.

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(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

(2)

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

(i) Describe a method for this investigation using the apparatus shown in the diagram.

(2)

(ii) Give two variables the student would need to control in their investigation.

(2)

1 .....

2 .....

(Total for Question 3 = 11 marks)

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5. Question 3 · part a.iv · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

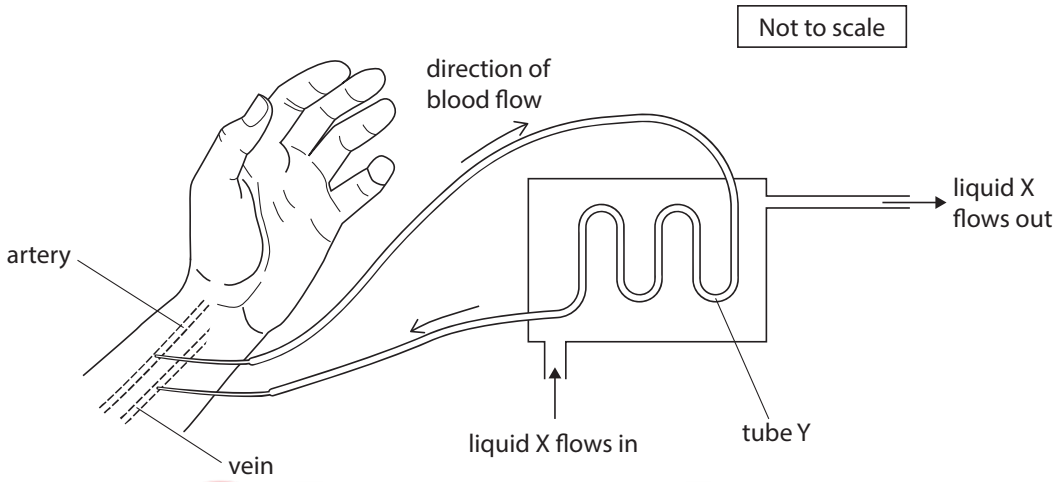
6. Question 5 · part a.iii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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6. Question 5 · part a.iii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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6. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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7. Question 7 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

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7. Question 7 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

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(iii) Calculate the mass, in grams, of boiled broccoli that should be eaten to provide the recommended daily intake of vitamin C. (lines 12 to 14)

(3)

mass = ... g

(b) DCPIP is a chemical used to determine the amount of vitamin C in food.

Describe a method, using DCPIP, to compare the amount of vitamin C in fresh orange juice with that in boiled orange juice.

(5)

(Total for Question 7 = 12 marks)

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7. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how inhibitors or a mutation change enzyme activity

Competitive and non-competitive inhibitors, drugs such as aspirin acting as inhibitors, and mutations that change an enzyme's shape.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-INHIBITOR-MUTATION

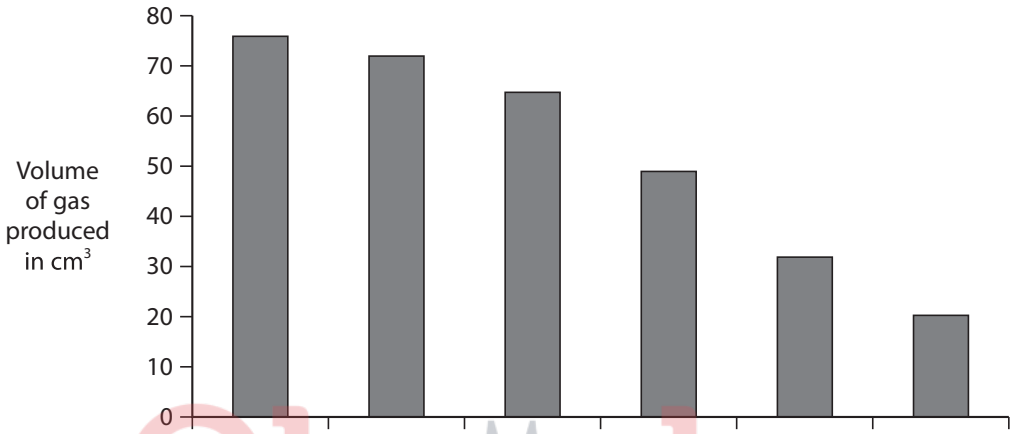
1. Question 3 · part b · 2 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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1. Question 3 · part b · 2 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

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(iv) Explain the student's results obtained in the investigation.

(3)

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(b) Explain the effect on the results if the student had added a competitive inhibitor at the start of the investigation.

(2)

(Total for Question 3 = 12 marks)

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1. Question 3 · part b · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b.iii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

2. Question 5 · part b.iii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

(iv) State why some mutations have no effect on the phenotype of an organism.

(1)

.....

.....

(Total for Question 5 = 9 marks)

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Working and answer

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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

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9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

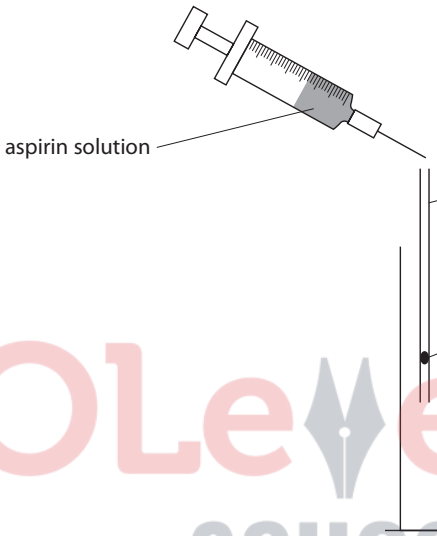
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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.



The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

(1)

(ii) Describe a suitable control test for this investigation.

(1)

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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

The table shows the student's results.

Time in minutes	Total volume of aspirin solution in the measuring cylinder in cm^3		
	1 aspirin tablet	2 aspirin tablets	3 aspirin tablets
5	0	0	0
10	0	0	0
15	0	0	1
20	0	1	2
25	1	1	3
30	1	2	4
35	2	4	6
40	5	7	9
45	10	12	16
50	17	19	22
55	25	27	31
60	35	37	39

(iii) Explain the pattern shown by the results.

(3)

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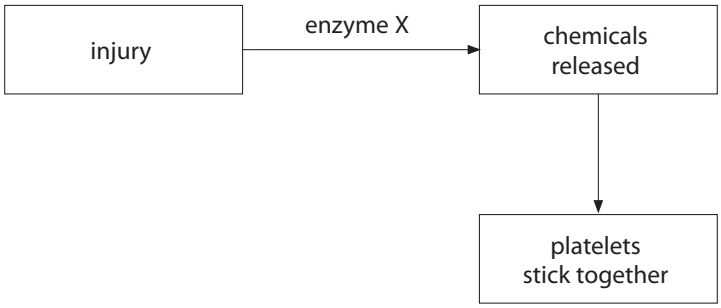
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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

(c) The diagram shows a pathway involved in blood clotting.



Aspirin is an inhibitor of enzyme X.

Explain how the inhibition of enzyme X reduces the formation of blood clots.

(4)

(Total for Question 9 = 15 marks)

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3. Question 9 · part c · 4 marks · Working and answer

Working and answer

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4. Question 2 · part b.iii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.iii only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

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2 .....

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3 .....

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4. Question 2 · part b.iii · 2 marks

June 2022 Paper 2R

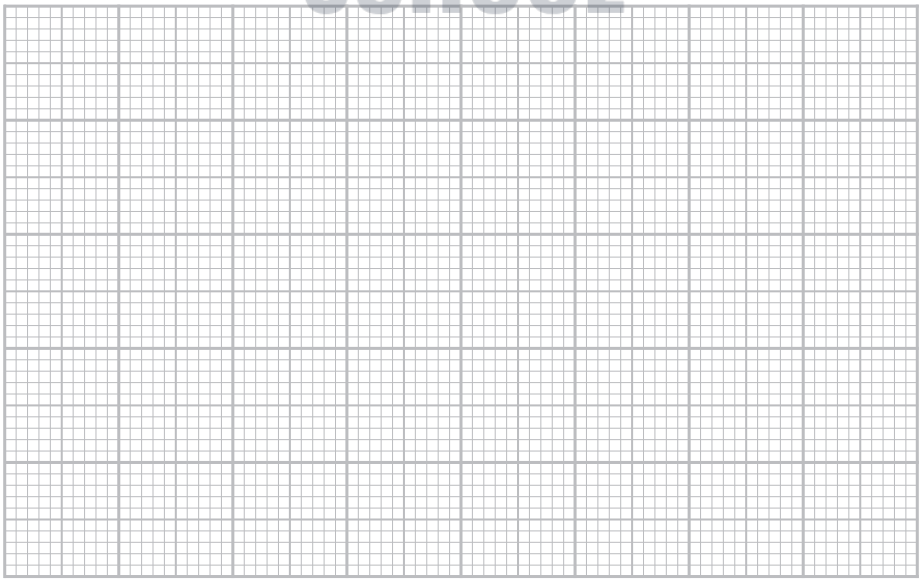
Further practice · Attempt part b.iii only.

(b) The table shows the student's results.

Concentration of phenolphthalein phosphate solution in arbitrary units	Intensity of pink colour in arbitrary units
0.05	0.20
0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**



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P 7 0 9 5 0 A 0 6 2 4

4. Question 2 · part b.iii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Add a second curve to show the results that would be obtained if the competitive inhibitor had not been added.

(2)

(c) Describe how the results of the investigation would be different if a non-competitive inhibitor had been used instead of a competitive inhibitor.

(2)

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(Total for Question 2 = 13 marks)

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Working and answer

5. Question 2 · part c · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

2 .....

3 .....

5. Question 2 · part c · 2 marks

June 2022 Paper 2R

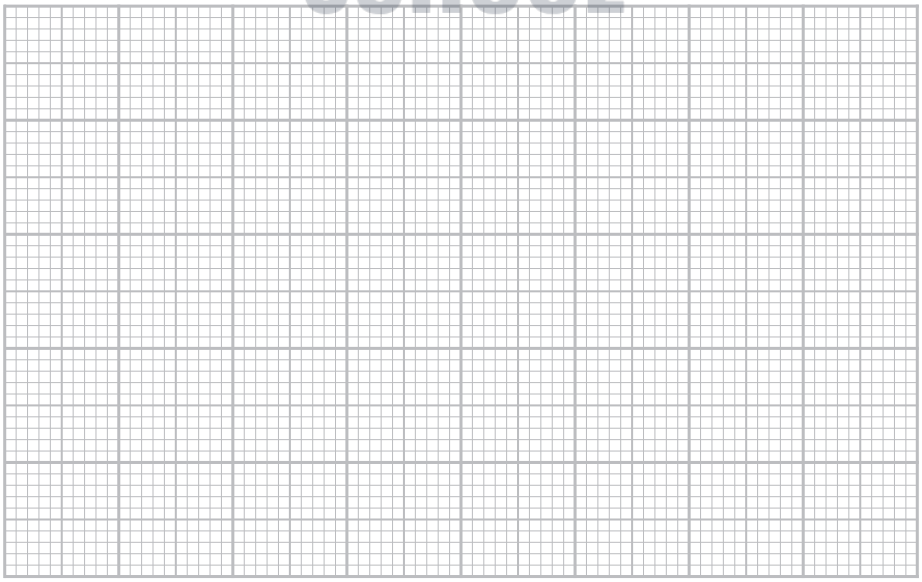
Further practice · Attempt part c only.

(b) The table shows the student's results.

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0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**



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5. Question 2 · part c · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

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(iii) Add a second curve to show the results that would be obtained if the competitive inhibitor had not been added.

(2)

(c) Describe how the results of the investigation would be different if a non-competitive inhibitor had been used instead of a competitive inhibitor.

(2)

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(Total for Question 2 = 13 marks)

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Working and answer

Practice workbook · Explain how temperature or pH changes enzyme activity

Explaining rising, falling or zero enzyme activity as temperature or pH changes, why tests are kept at 37 °C or one pH, and why body temperature and pH must stay near normal.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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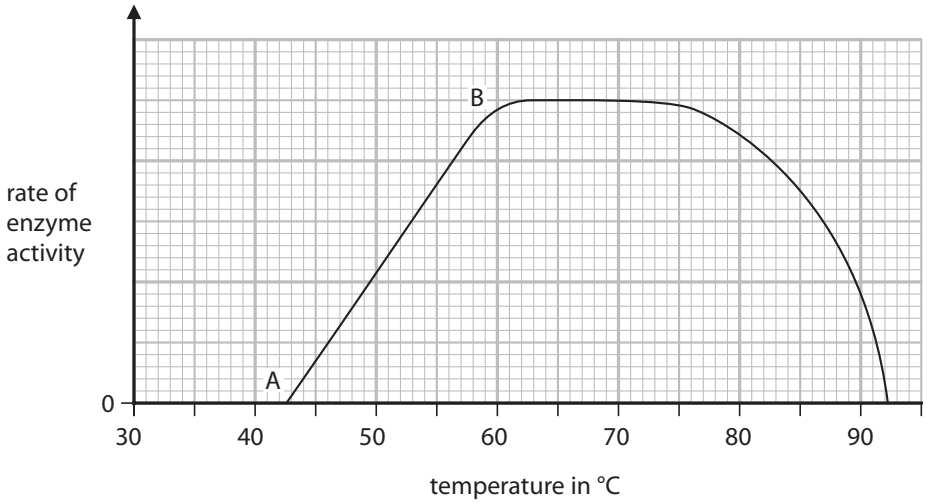
Type ID: HBI-ENZYME-TEMP-PH-DENATURE

1. Question 4 · part b · 5 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b only.

4 The graph shows the effect of temperature on the activity of a bacterial enzyme.



(a) Draw a curve on the grid to show the rate of enzyme activity for an enzyme from the human alimentary canal.

(2)

(b) Explain what is causing the change in enzyme activity from A to B.

(5)

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1. Question 4 · part b · 5 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b only.

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(c) Deduce which type of environment the bacterium lives in.

(2)

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(Total for Question 4 = 9 marks)

Working and answer

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2. Question 3 · part b.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.ii only.

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3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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2. Question 3 · part b.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.ii only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

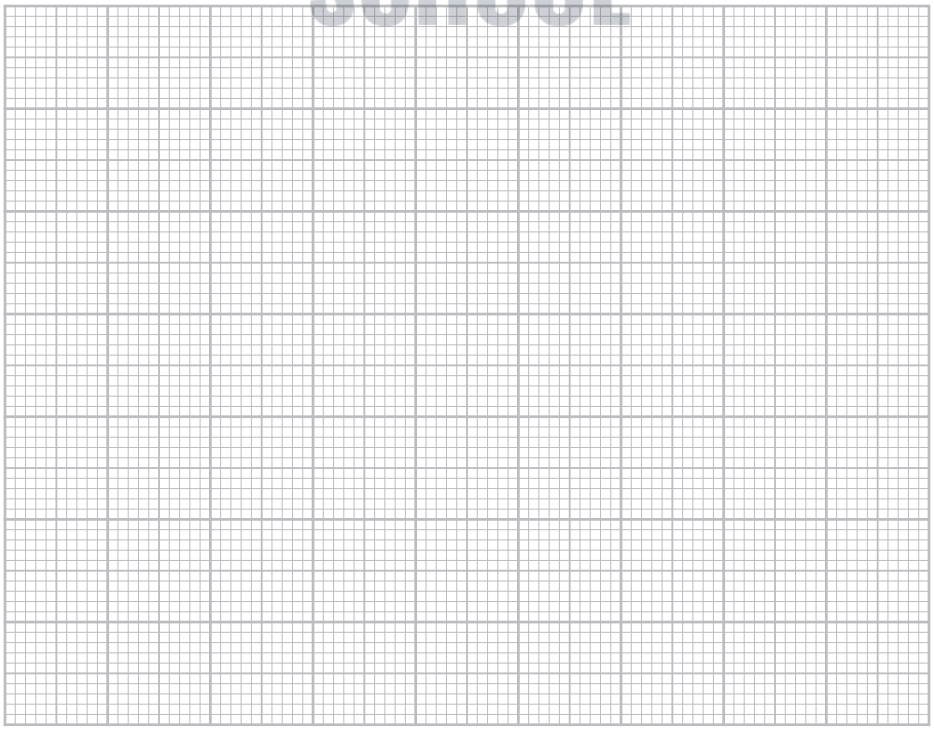
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



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P 6 6 4 3 2 A 0 8 2 4

2. Question 3 · part b.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Describe the effects on enzyme B of increasing the temperature from 80°C to 90°C.

(3)

(iii) Describe the differences between the rate of protein digestion shown by enzyme A and enzyme B.

Refer to the graph in your answer.

(3)

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(Total for Question 3 = 15 marks)

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2. Question 3 · part b.ii · 3 marks · Working and answer

Working and answer

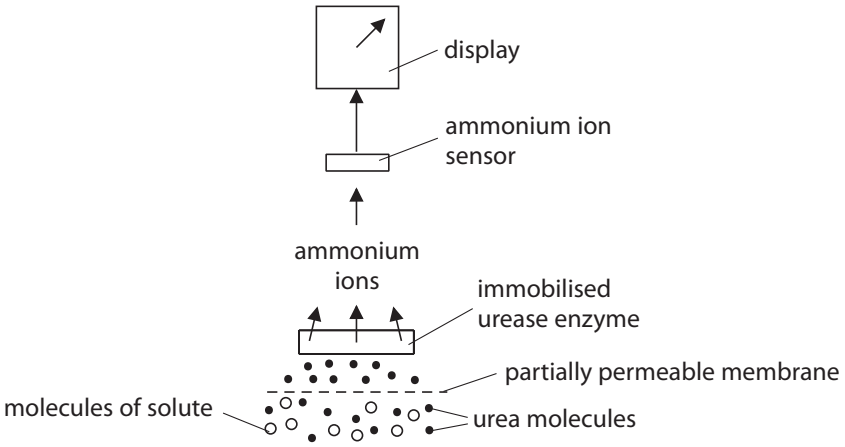
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3. Question 3 · part b · 4 marks

June 2021 Paper 2

Further practice · Attempt part b only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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3. Question 3 · part b · 4 marks

June 2021 Paper 2

Further practice · Attempt part b only.

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(b) A scientist wants to use the biosensor to compare the urea concentration of two urine samples.

Explain why it is important to do the tests at the same temperature.

(4)

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(c) Immobilised enzymes are also used in industry to produce useful molecules.

Give two advantages of using immobilised enzymes instead of enzymes that are free to move in solution.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

3. Question 3 · part b · 4 marks · Working and answer

Working and answer

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4. Question 4 · part b.iii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

4 (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

(3)

(ii) Name the enzyme that digests lipids.

(1)

(iii) Name the products of lipid digestion.

(2)

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

Temperature at 37 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	8.5	8.0	7.5	7.2	7.0	6.8	6.2	5.8	5.5	5.5

Temperature at 80 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	9.2	9.0	8.8	8.8		8.6	8.6	8.4	8.4	8.4

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4. Question 4 · part b.iii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

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(i) Estimate a pH value for the lipid solution at 80 °C at 5 minutes.

(1)

(ii) Give the independent variable and the dependent variable for this investigation.

(2)

independent

dependent

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(iii) Explain the differences in the results of the student's investigation at the two different temperatures.

(4)

(Total for Question 4 = 13 marks)

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4. Question 4 · part b.iii · 4 marks · Working and answer

Working and answer

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5. Question 5 · part b · 4 marks

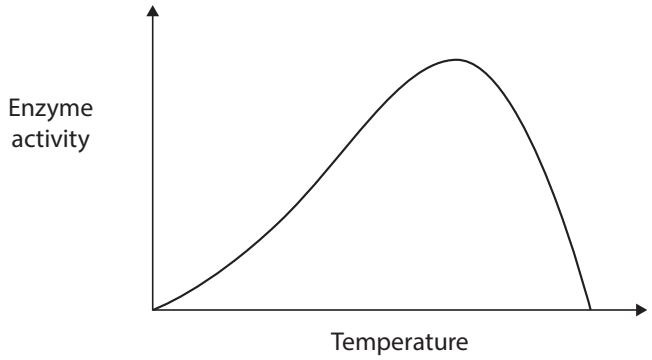
June 2024 Paper 1

Further practice · Attempt part b only.

5 (a) A group of students investigate the effect of temperature on enzyme activity.

The students use the results they obtain to plot a graph.

The graph shows the students' results.



(Source: adapted from https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/25-enzymes/enzyme-activity.html)

(i) Design a method the students could use to obtain the results shown in the graph.

(5)

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P 7 4 6 2 6 R A 0 1 2 2 4

5. Question 5 · part b · 4 marks

June 2024 Paper 1

Further practice · Attempt part b only.

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(ii) State the independent variable in this investigation.

(1)

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(b) Explain how enzyme activity is affected by low temperatures.

(4)

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(Total for Question 5 = 10 marks)

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5. Question 5 · part b · 4 marks · Working and answer

Working and answer

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6. Question 8 · part c · 5 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

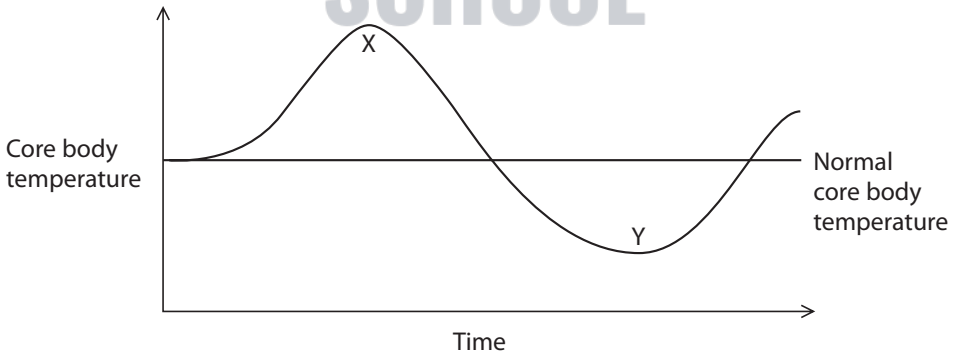
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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6. Question 8 · part c · 5 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

(iii) Explain why the control of body temperature is an example of negative feedback.

(2)

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6. Question 8 · part c · 5 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

(c) Hypothermia is a condition where the core body temperature drops below 35 °C.

Explain why it is important to increase the core body temperature of people with hypothermia to within the normal range.

(5)

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(Total for Question 8 = 14 marks)

6. Question 8 · part c · 5 marks · Working and answer

Working and answer

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7. Question 7 · part a · 2 marks

November 2024 Paper 2

Further practice · Attempt part a only.

7 Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

Student	Volume of enzyme solution in cm^3	Temperature in °C	pH	Volume of cloudy suspension in cm^3	Time tubes left in minutes	Appearance at end of investigation
1	5	37	2	10	9	cloudy
2	5	37	2	10	20	clear
3	5	37	4	10	35	clear
4	5	37	4	10	20	cloudy
5	5	37	6	10	8	cloudy
6	5	37	6	10	40	cloudy

(a) Explain why the investigation was carried out at 37 °C.

(2)

(b) Explain which property of the enzyme the students are investigating.

(2)

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7. Question 7 · part a · 2 marks · Working and answer

Working and answer

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8. Question 3 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

3 The control of body temperature is achieved by a process known as homeostasis.

(a) Explain what is meant by homeostasis.

(4)

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8. Question 3 · part c · 4 marks

June 2025 Paper 1

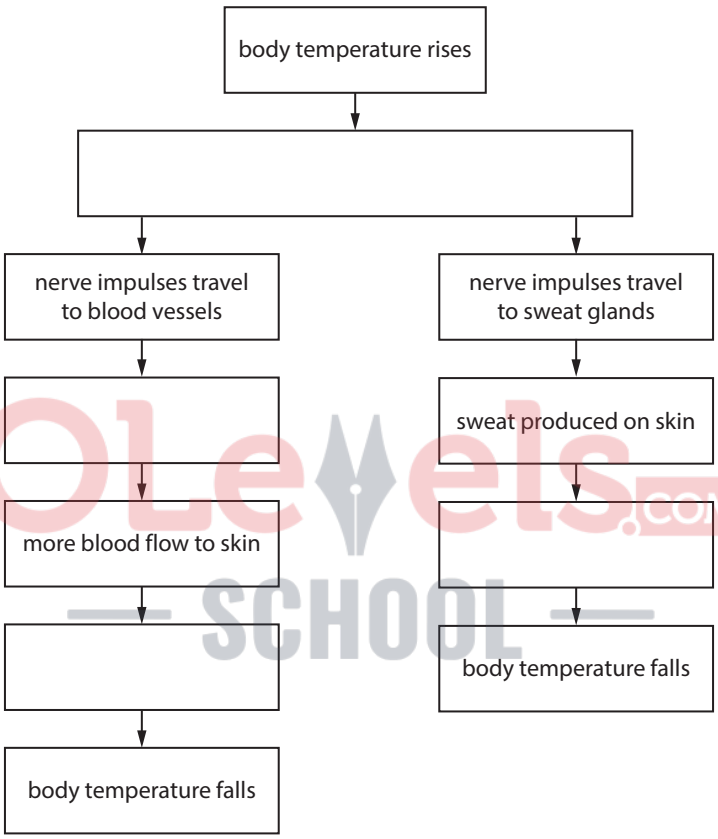
Further practice · Attempt part c only.

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(b) The diagram shows some of the reactions of the body to an increase in temperature.



Complete the diagram by putting in a suitable word or sentence in each of the blank boxes.

(6)

P 7 8 7 6 8 A 0 7 2 0

Turn over ▶

8. Question 3 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

(c) Explain why it is important that body temperature should not become too high.

(4)

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(Total for Question 3 = 14 marks)

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Working and answer

9. Question 2 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

2 A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

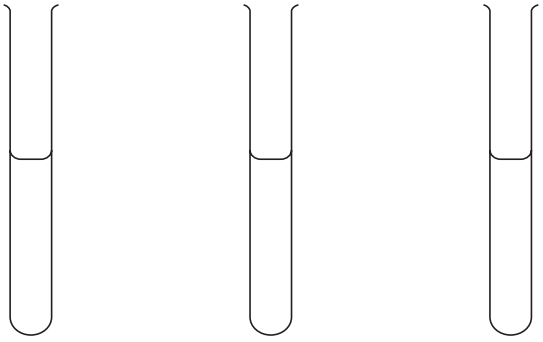


Table 1 lists the contents of each test tube.

Test tube A	Test tube B	Test tube C
5 cm^3 of milk	5 cm^3 of milk	5 cm^3 of milk
2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate
6 drops of phenolphthalein	6 drops of phenolphthalein	6 drops of phenolphthalein
1 cm^3 enzyme	1 cm^3 enzyme	1 cm^3 boiled enzyme
distilled water	bile salts	bile salts

Table 1

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9. Question 2 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

	Colour of contents		
	Tube A	**Tube B**	**Tube C**
at start	pink	pink	pink
after 5 minutes	pink	colourless	pink
after 10 minutes	pink	colourless	pink
after 15 minutes	colourless	colourless	pink

Table 2

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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9. Question 2 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

(b) Explain the purpose of tube C.

(2)

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

(2)

(d) Describe the effect of bile salts on lipid digestion in this investigation.

(3)

(Total for Question 2 = 10 marks)

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9. Question 2 · part c · 2 marks · Working and answer

Working and answer

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10. Question 1 · part b.iii · 4 marks

November 2021 Paper 1

Further practice · Attempt part b.iii only.

1 The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 cm^3 of water, and mix into a smooth paste
- put 10 cm^3 of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

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10. Question 1 · part b.iii · 4 marks

November 2021 Paper 1

Further practice · Attempt part b.iii only.

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(a) (i) Give the student's results in the form of a suitable table.

(3)

(ii) State why this investigation is not reliable.

(1)

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(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

10. Question 1 · part b.iii · 4 marks

November 2021 Paper 1

Further practice · Attempt part b.iii only.

(b) (i) State the condition where the enzyme was most active.

(1)

(ii) State the reason for including the beaker with no hydrogen peroxide added.

(1)

(iii) Explain the result obtained when the conditions were acidic.

(4)

(Total for Question 1 = 14 marks)

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10. Question 1 · part b.iii · 4 marks · Working and answer

Working and answer

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11. Question 6 · part b · 2 marks

January 2023 Paper 1

Further practice · Attempt part b only.

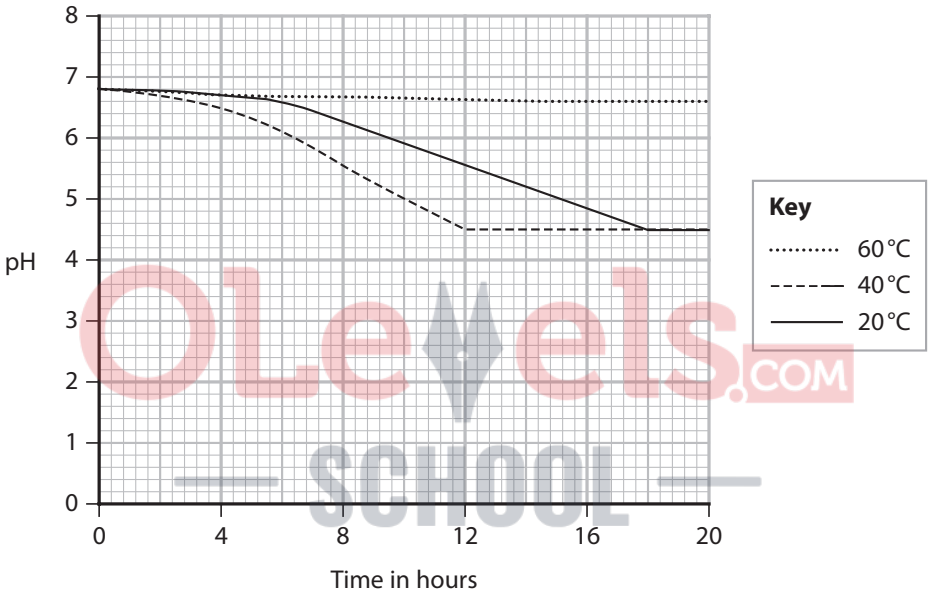
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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11. Question 6 · part b · 2 marks

January 2023 Paper 1

Further practice · Attempt part b only.

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(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

.....

.....

cooling

.....

.....

11. Question 6 · part b · 2 marks

January 2023 Paper 1

Further practice · Attempt part b only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

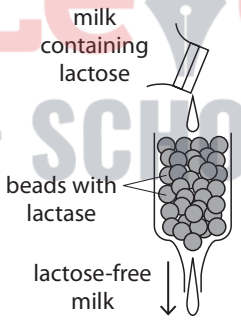
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

(2)

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.



(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

(2)

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P 7 2 2 7 1 A 0 1 8 2 4

11. Question 6 · part b · 2 marks · Working and answer

Working and answer

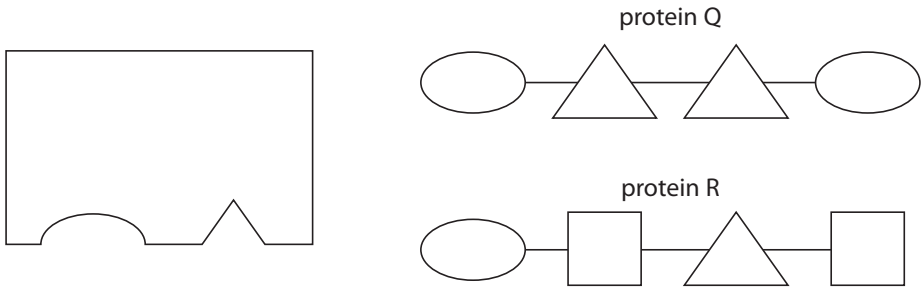
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12. Question 2 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

2 (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.



Enzyme P **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

(3)

amino acids	amylases	fatty acids	proteases
protein Q	protein R	starch	sugar

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

(1)

.....

(iii) Name the molecule that codes for the production of enzyme P.

(1)

.....

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12. Question 2 · part b.ii · 2 marks

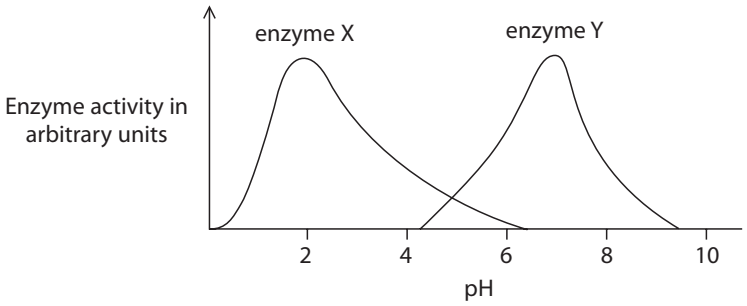
June 2023 Paper 2R

Further practice · Attempt part b.ii only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.



(i) Describe how the activity of enzyme X changes with pH.

(3)

(ii) Explain why no activity is shown for enzyme Y at pH2.

(2)

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P 7 2 5 8 8 A 0 6 2 4

12. Question 2 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Describe how the student could make the results of this investigation more reliable.

(2)

.....

.....

.....

.....

(iv) Name one control variable in the student's investigation.

(1)

.....

(Total for Question 2 = 13 marks)

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Working and answer

13. Question 4 · part a.v · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

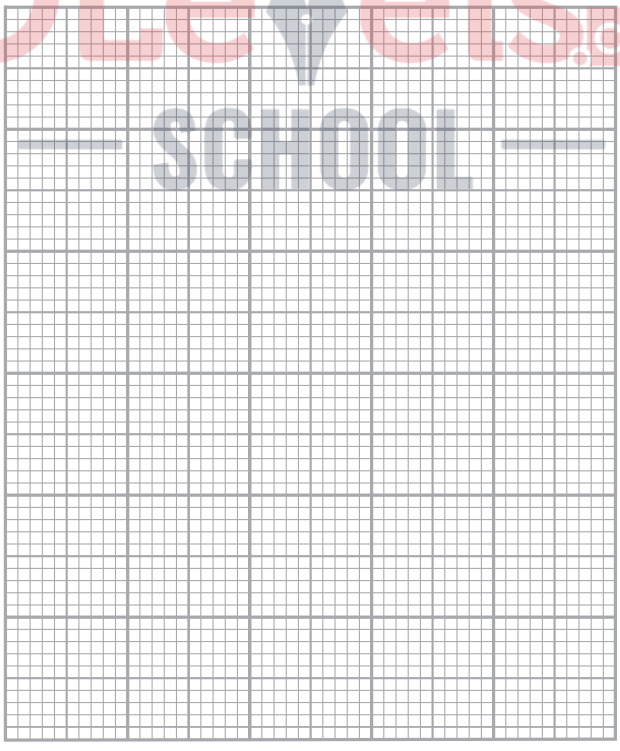
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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13. Question 4 · part a.v · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

(Total for Question 4 = 13 marks)

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13. Question 4 · part a.v · 3 marks · Working and answer

Working and answer

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14. Question 4 · part b.ii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part b.ii only.

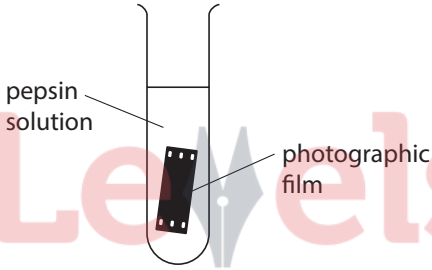
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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14. Question 4 · part b.ii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part b.ii only.

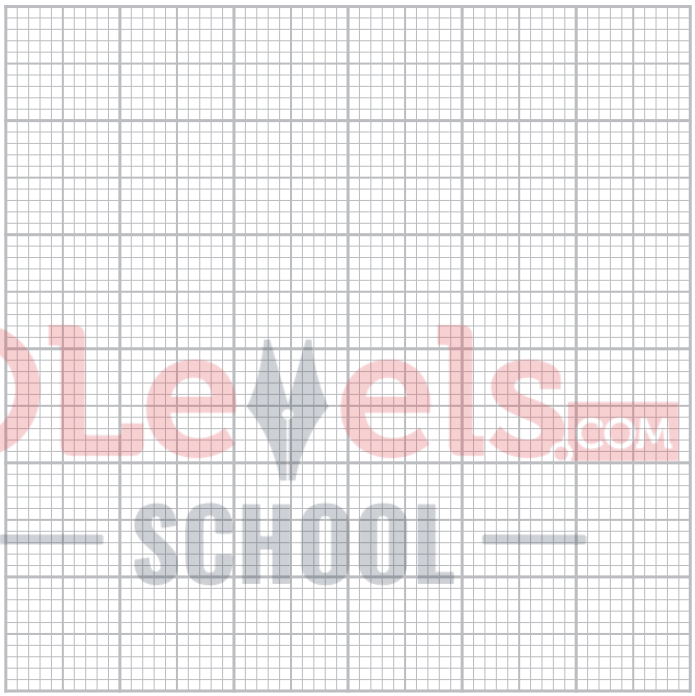
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(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



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(b) (i) Explain why the photographic film becomes clear.

(3)

14. Question 4 · part b.ii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part b.ii only.

(ii) Explain which would be the best pH to carry out this investigation.

(2)

(iii) State three variables that should be controlled during this investigation other than pH.

(3)

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

(2)

(Total for Question 4 = 15 marks)

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14. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Read an enzyme optimum from data and judge how reliable it is

Tables, bar charts or graphs of enzyme activity against temperature or pH: reading or sketching the optimum, comparing enzymes, concluding, and judging how well the data pin down the optimum.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-OPTIMUM-FROM-DATA

1. Question 4 · part a.iii · 1 mark

June 2024 Paper 1R

Worked-example retry · Attempt part a.iii only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

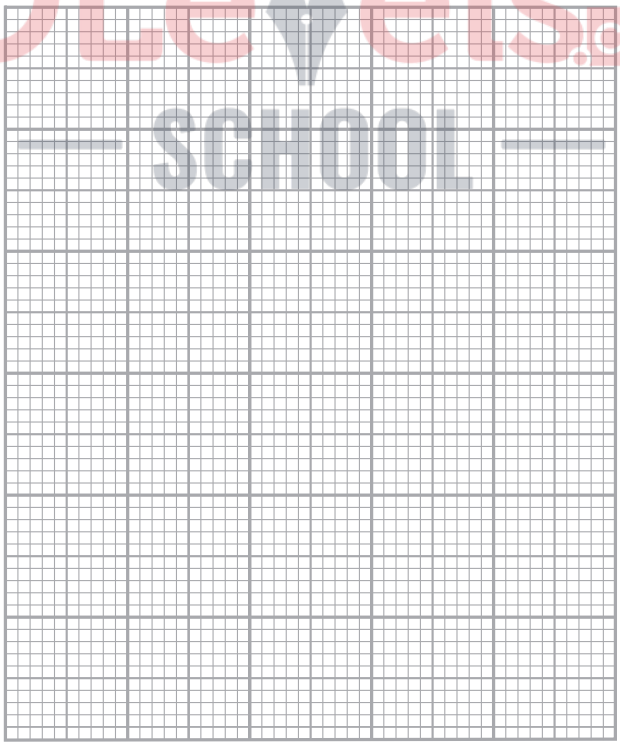
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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1. Question 4 · part a.iii · 1 mark

June 2024 Paper 1R

Worked-example retry · Attempt part a.iii only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

(Total for Question 4 = 13 marks)

1. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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2. Question 7 · part b.i · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b.i only.

7 Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

Time after adding enzyme in hours	Amount of lactic acid in the solution in arbitrary units		
	2 °C	25 °C	55 °C
0	0.1	0.1	0.1
10	0.1	0.2	0.2
20	0.1	0.3	0.2
30	0.2	0.8	0.4
40	0.3	1.8	0.7
50	0.4	2.2	1.0
60	0.6	2.3	1.3

(a) Describe what additional information another student would need to repeat this investigation.

(4)

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2. Question 7 · part b.i · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b.i only.

(b) (i) Estimate the optimum temperature for this enzyme.

(1)

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

(2)

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

(3)

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2. Question 7 · part b.i · 1 mark · Working and answer

Working and answer

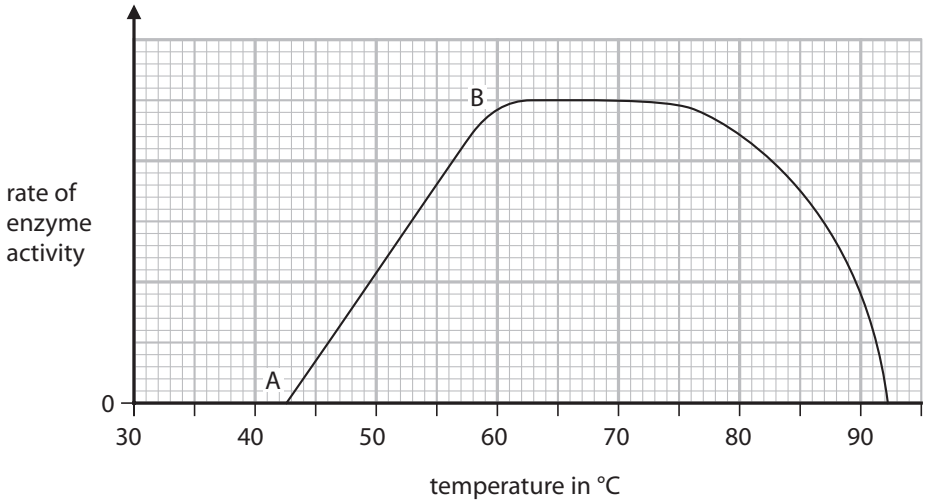
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3. Question 4 · part a · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a only.

4 The graph shows the effect of temperature on the activity of a bacterial enzyme.



(a) Draw a curve on the grid to show the rate of enzyme activity for an enzyme from the human alimentary canal.

(2)

(b) Explain what is causing the change in enzyme activity from A to B.

(5)

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3. Question 4 · part a · 2 marks · Working and answer

Working and answer

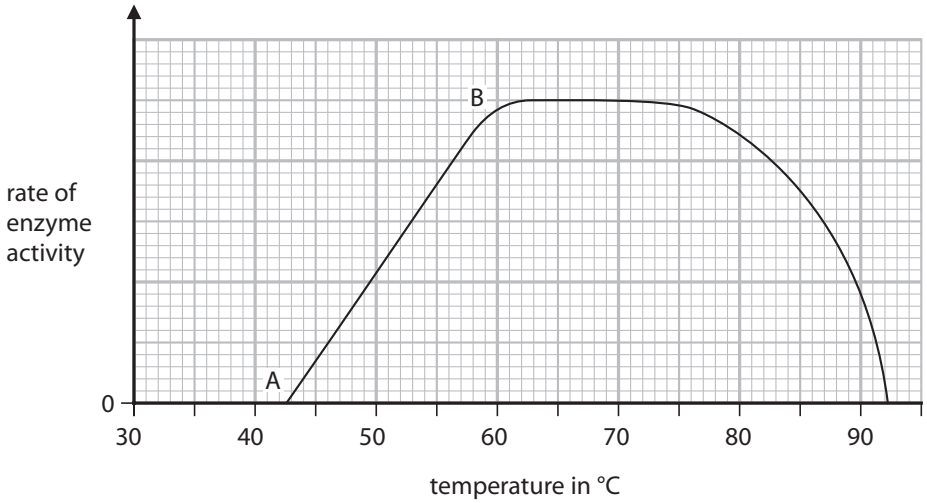
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4. Question 4 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

4 The graph shows the effect of temperature on the activity of a bacterial enzyme.



(a) Draw a curve on the grid to show the rate of enzyme activity for an enzyme from the human alimentary canal.

(2)

(b) Explain what is causing the change in enzyme activity from A to B.

(5)

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4. Question 4 · part c · 2 marks

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Further practice · Attempt part c only.

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(c) Deduce which type of environment the bacterium lives in.

(2)

.....

.....

.....

.....

(Total for Question 4 = 9 marks)

Working and answer

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5. Question 3 · part b.iii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.iii only.

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3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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5. Question 3 · part b.iii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.iii only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

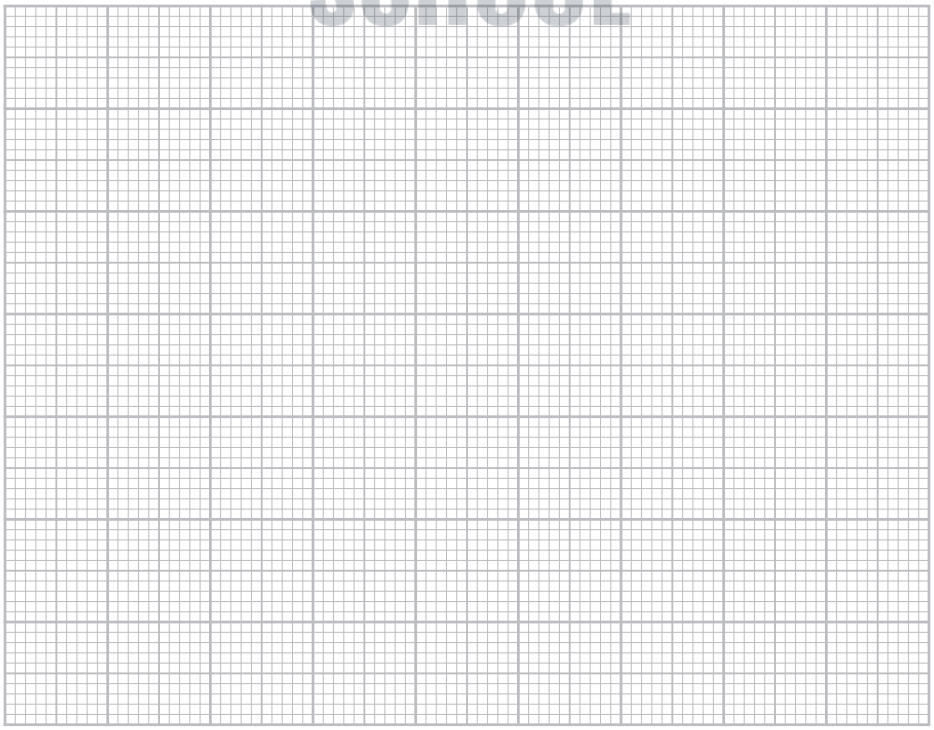
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



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P 6 6 4 3 2 A 0 8 2 4

5. Question 3 · part b.iii · 3 marks

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Further practice · Attempt part b.iii only.

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(ii) Describe the effects on enzyme B of increasing the temperature from 80°C to 90°C.

(3)

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(iii) Describe the differences between the rate of protein digestion shown by enzyme A and enzyme B.

Refer to the graph in your answer.

(3)

(Total for Question 3 = 15 marks)

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5. Question 3 · part b.iii · 3 marks · Working and answer

Working and answer

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6. Question 1 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

1 The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 cm^3 of water, and mix into a smooth paste
- put 10 cm^3 of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

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6. Question 1 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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(a) (i) Give the student's results in the form of a suitable table.

(3)

(ii) State why this investigation is not reliable.

(1)

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(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

6. Question 1 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

(b) (i) State the condition where the enzyme was most active.

(1)

(ii) State the reason for including the beaker with no hydrogen peroxide added.

(1)

(iii) Explain the result obtained when the conditions were acidic.

(4)

(Total for Question 1 = 14 marks)

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Working and answer

7. Question 7 · part a.ii · 1 mark

November 2023 Paper 2

Further practice · Attempt part a.ii only.

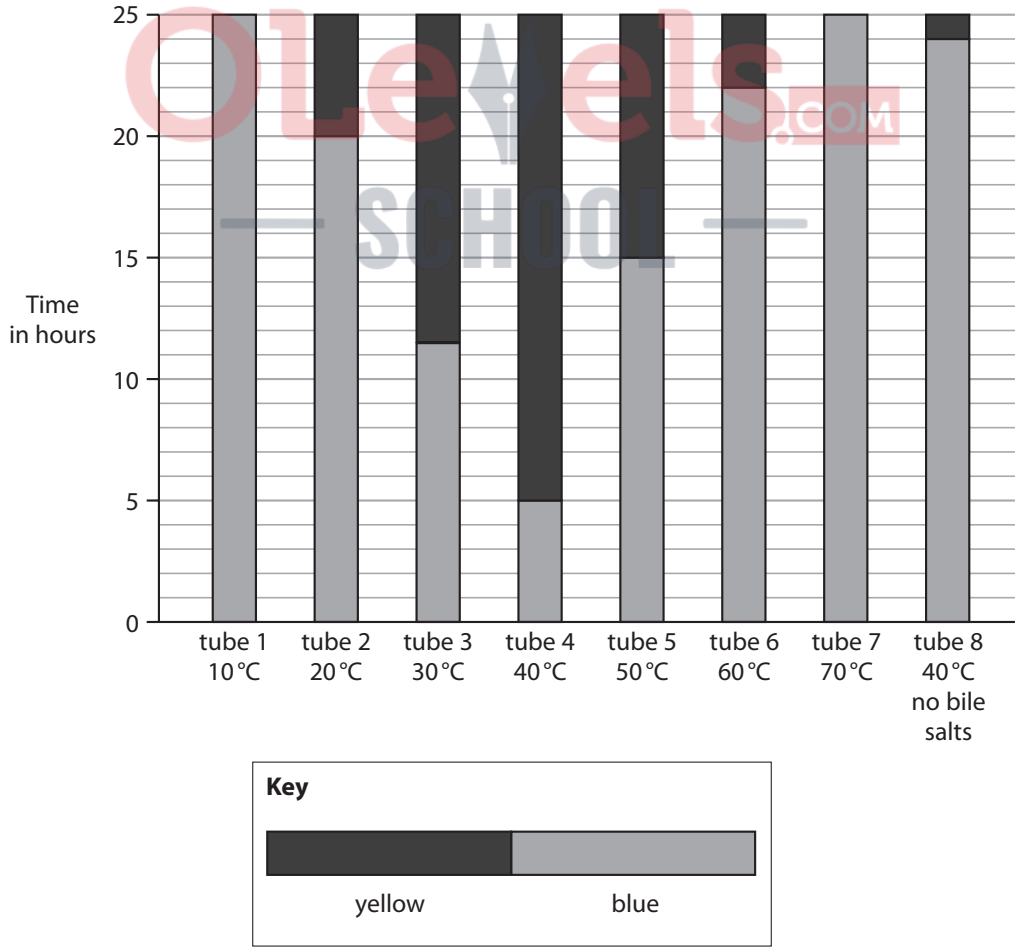
7 A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.



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7. Question 7 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

(3)

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

(1)

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

(3)

.....

.....

.....

.....

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.....

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

(2)

time = ... hours

(ii) Explain why the pH decreases during the 25 hours.

(3)

.....

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7. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 7 · part e · 2 marks

November 2024 Paper 2

Further practice · Attempt part e only.

7 Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

Student	Volume of enzyme solution in cm^3	Temperature in °C	pH	Volume of cloudy suspension in cm^3	Time tubes left in minutes	Appearance at end of investigation
1	5	37	2	10	9	cloudy
2	5	37	2	10	20	clear
3	5	37	4	10	35	clear
4	5	37	4	10	20	cloudy
5	5	37	6	10	8	cloudy
6	5	37	6	10	40	cloudy

(a) Explain why the investigation was carried out at 37 °C.

(2)

(b) Explain which property of the enzyme the students are investigating.

(2)

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8. Question 7 · part e · 2 marks

November 2024 Paper 2

Further practice · Attempt part e only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

(4)

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(d) Describe a control experiment that the students could have carried out.

(3)

8. Question 7 · part e · 2 marks

November 2024 Paper 2

Further practice · Attempt part e only.

(e) State a possible conclusion that the students could make from the results of this investigation.

(2)

(f) Explain where this enzyme is likely to be found in the alimentary canal.

(2)

(Total for Question 7 = 15 marks)

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Working and answer

9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

7 Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

Time after adding enzyme in hours	Amount of lactic acid in the solution in arbitrary units		
	2 °C	25 °C	55 °C
0	0.1	0.1	0.1
10	0.1	0.2	0.2
20	0.1	0.3	0.2
30	0.2	0.8	0.4
40	0.3	1.8	0.7
50	0.4	2.2	1.0
60	0.6	2.3	1.3

(a) Describe what additional information another student would need to repeat this investigation.

(4)

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9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(b) (i) Estimate the optimum temperature for this enzyme.

(1)

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

(2)

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

(3)

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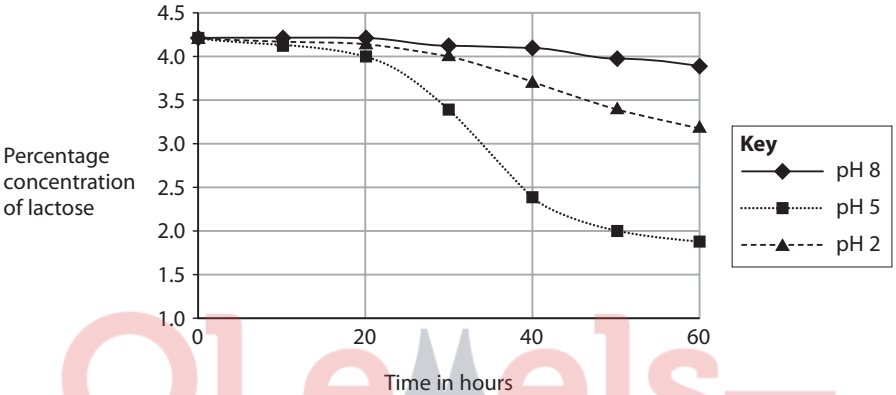
9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(d) The student uses the same enzyme to investigate the effect of pH on the conversion of the lactose into lactic acid.

The student's results are shown in the graph.



The student wants to determine the most effective conditions for the conversion of lactose into lactic acid in cheese manufacture.

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9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

Discuss if this is possible using the results of the two investigations.

(5)

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(Total for Question 7 = 15 marks)

9. Question 7 · part d · 5 marks · Working and answer

Working and answer

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10. Question 7 · part a.iii · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

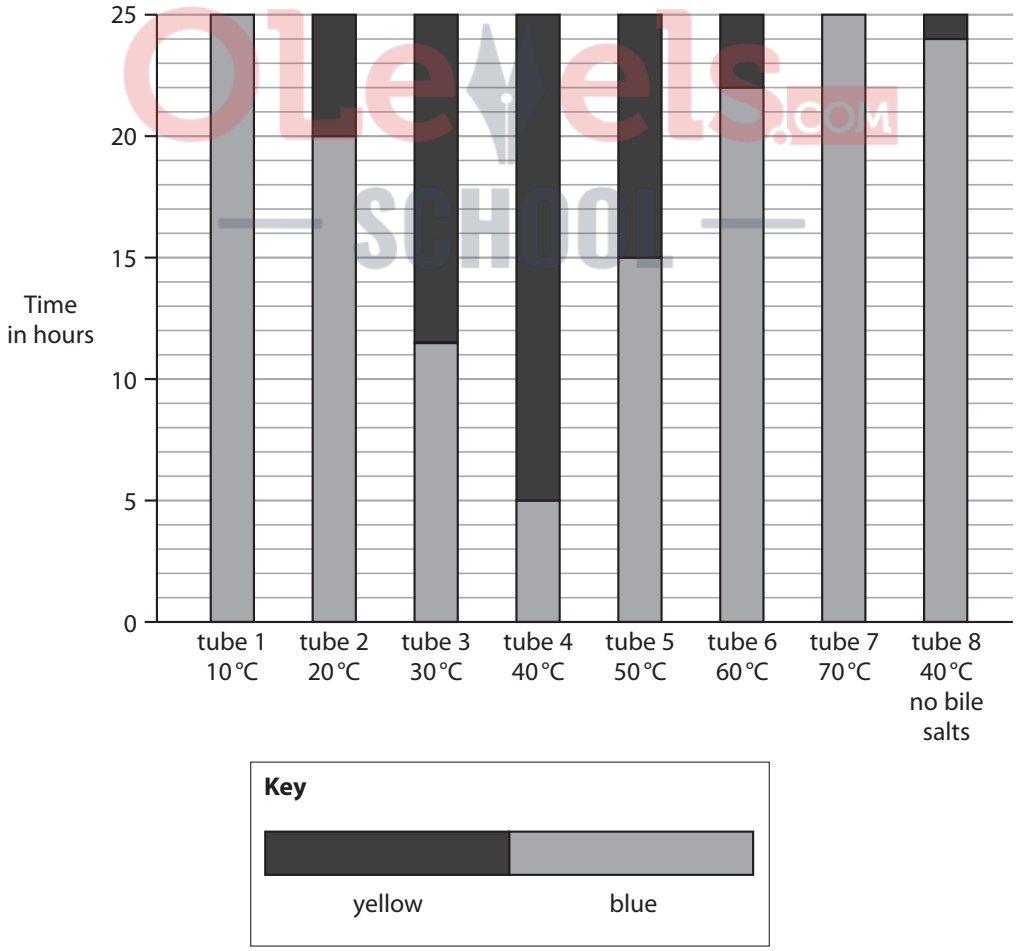
7 A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.



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10. Question 7 · part a.iii · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

(3)

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

(1)

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

(3)

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

(2)

time = ... hours

(ii) Explain why the pH decreases during the 25 hours.

(3)

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10. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

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11. Question 7 · part c · 4 marks

November 2024 Paper 2

Further practice · Attempt part c only.

7 Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

Student	Volume of enzyme solution in cm^3	Temperature in °C	pH	Volume of cloudy suspension in cm^3	Time tubes left in minutes	Appearance at end of investigation
1	5	37	2	10	9	cloudy
2	5	37	2	10	20	clear
3	5	37	4	10	35	clear
4	5	37	4	10	20	cloudy
5	5	37	6	10	8	cloudy
6	5	37	6	10	40	cloudy

(a) Explain why the investigation was carried out at 37 °C.

(2)

(b) Explain which property of the enzyme the students are investigating.

(2)

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11. Question 7 · part c · 4 marks

November 2024 Paper 2

Further practice · Attempt part c only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

(4)

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(d) Describe a control experiment that the students could have carried out.

(3)

11. Question 7 · part c · 4 marks · Working and answer

Working and answer

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Practice workbook · Explain water movement by osmosis in body contexts

Applying water potential to blood, cells, gut and tissue fluid in diabetes, dehydration, lactose intolerance and oedema, plus finding cell concentration from osmosis data.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-OSMOSIS-IN-BODY

1. Question 8 · part c · 3 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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1. Question 8 · part c · 3 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

(d) Suggest the effect that cellular oedema has on body cells.

(2)

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(Total for Question 8 = 11 marks)

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Working and answer

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2. Question 5 · part c.iii · 4 marks

June 2019 Paper 2R

Further practice · Attempt part c.iii only.

5 Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

(2)

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

(5)

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2. Question 5 · part c.iii · 4 marks

June 2019 Paper 2R

Further practice · Attempt part c.iii only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

	Concentration of sucrose solution in mol per dm³				
	0.2	**0.4**	**0.6**	**0.8**	**1.0**
Initial mass in g	8.4	7.9	8.7	8.3	8.4
Mass after soaking in g	8.7	8.0	8.2	7.5	7.3
Change in mass in g	+ 0.3	+ 0.1	– 0.5	– 0.8	– 1.1
Percentage change in mass	+ 3.6		– 5.8	– 9.6	– 13.1

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

(2)

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

(2)

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2. Question 5 · part c.iii · 4 marks

June 2019 Paper 2R

Further practice · Attempt part c.iii only.

(iii) Explain how the results in the table could be used to determine the concentration of the contents of potato cells.

(4)

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(Total for Question 5 = 15 marks)

Working and answer

3. Question 7 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

.....

2 .....

.....

(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

.....

2 .....

.....

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3. Question 7 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) (i) Individuals who are overweight or obese have an increased risk of diabetes.

People with diabetes can have high blood glucose levels.

Explain how blood glucose levels are reduced in a person who does not have diabetes.

(3)

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(ii) Suggest why one symptom of diabetes is frequent urination.

(2)

(Total for Question 7 = 11 marks)

3. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 6 · part d · 2 marks

November 2020 Paper 2

Further practice · Attempt part d only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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4. Question 6 · part d · 2 marks

November 2020 Paper 2

Further practice · Attempt part d only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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4. Question 6 · part d · 2 marks

November 2020 Paper 2

Further practice · Attempt part d only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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4. Question 6 · part d · 2 marks · Working and answer

Working and answer

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5. Question 5 · part e · 4 marks

June 2022 Paper 2R

Further practice · Attempt part e only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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5. Question 5 · part e · 4 marks

June 2022 Paper 2R

Further practice · Attempt part e only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

5. Question 5 · part e · 4 marks · Working and answer

Working and answer

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6. Question 8 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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6. Question 8 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

(d) Suggest the effect that cellular oedema has on body cells.

(2)

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(Total for Question 8 = 11 marks)

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Working and answer

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